

Series

Unit 1-4 Lecture Notes

MSDS 520: Math Fundamentals for Data Science
Meharry Medical College

These notes are the written companion to the Unit 1-4 slides and follow them section by section. Every exercise is followed by a worked solution; try each one before reading on.

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1 Why Series?

1.1 A Problem to Motivate Us

Here is a question a pharmacist or a clinician might actually ask:

“A patient takes 100 mg of a drug every 6 hours. The drug’s half-life is also 6 hours, so half of what is in the body is gone by the time the next dose arrives. Does the drug pile up forever?”

The **half-life** of a drug is the time it takes the body to clear half of whatever amount is present. Because the half-life here matches the dosing interval, every dose already in the body has been cut in half by the time the next one is taken.

A word about the model before we use it. We treat each dose as entering the body all at once, and we assume the body clears a fixed *fraction* of whatever is present over each interval, which is exactly what a constant half-life means. Real pharmacology adds refinements (a pill takes time to

be absorbed, for instance), but this simple version already captures the main behavior, and it is the model used throughout these notes.

Track the amount in the body just after each dose. After the first dose there are 100 mg. By the time of the second dose, that first 100 mg has halved to $100(0.5)$, and a fresh 100 mg arrives on top of it. By the third dose, the first dose has halved twice, to $100(0.5)^2$, the second has halved once, and a new 100 mg arrives. The pattern continues:

$$\begin{aligned}\text{after dose 1: } & 100 \\ \text{after dose 2: } & 100 + 100(0.5) \\ \text{after dose 3: } & 100 + 100(0.5) + 100(0.5)^2 \\ \text{after dose } n : & 100 + 100(0.5) + 100(0.5)^2 + \cdots + 100(0.5)^{n-1}\end{aligned}$$

It helps to put actual numbers on this before doing any theory. Adding up the terms for the first six doses:

Dose number	Amount just after this dose (mg)	Increase since the previous dose
1	100	(first dose)
2	150	50
3	175	25
4	187.5	12.5
5	193.75	6.25
6	196.875	3.125

Two things are visible in the table. The amount goes up after every dose, so the drug certainly does accumulate. But each increase is exactly half the one before it, so the growth is slowing down. A quantity that always grows, but by less and less each time, could in principle go either way: it might creep upward forever, or it might level off below some ceiling. Telling those two cases apart is exactly what the mathematics of series is for.

Every dose adds another term, so the sum has more in it every day. It is tempting to conclude that the total must keep growing without limit, and yet the answer is **no**: the total does not grow forever. Understanding *why* a sum with more and more terms can still stay below a fixed ceiling is what these notes are about, and we answer the dosing question exactly in Section 5, along with several variations on it (other dose sizes, other half-lives, and how many doses it takes to get close to the ceiling).

The same puzzle hides inside a fact you may have met in school, that the endless decimal $0.999\ldots$ equals 1. That decimal *means* $0.9 + 0.09 + 0.009 + \cdots$, a sum with infinitely many positive terms, and many people feel it must fall “just short” of 1. Section 4.2 settles it with the same tool that settles the dosing question.

1.2 Where This Shows Up in Data Science

You are already summing things constantly, even if you have never written it down in symbols:

- **Every statistic is a sum:** a mean, a variance, a total count. Each one adds something up over the rows of a data set. The mean fasting glucose of five patients, for instance, is the total of their five readings divided by 5.
- **Every loss function is a sum:** a *loss function* measures how wrong a model is, and it is built by adding up how wrong the model was on each of the n training rows.
- **Every dot product is a sum:** multiply two lists of numbers entry by entry and add the results. This is the core operation of Unit 2, and of every neural network. A pharmacy bill is one: quantity times unit price for each item, then add.
- **Computers only add:** e^x , $\sin x$ and $\ln x$ are all evaluated as sums of infinitely many simple terms, cut off partway. For example, $1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} + \frac{1}{120} \approx 2.7167$ is already close to $e \approx 2.71828$, and adding more terms of the same kind closes the gap. (Unit 3 builds those sums and explains where the fractions come from.)
- **Training is a sequence of steps:** a training algorithm takes one small step after another, so where it stands after n steps is the start plus the sum of the first n steps. Asking “does this algorithm settle down?” is exactly the question of whether a series converges. If the steps were 1, then $\frac{1}{2}$, then $\frac{1}{4}$, and so on, the total distance traveled would close in on 2 without ever passing it; Section 4.1 plots exactly that series.

The first three are the reason **sigma notation** is worth the time we spend on it: it is the language all of them are written in.

This topic also picks up where the last one left off. Unit 1-3 built **functions**, rules that take an input and return an output. A series is what you get when you feed a function the numbers $1, 2, 3, \dots$ and add up the results.

1.3 What We Will Build

Over the rest of these notes we build up the tools in this order:

1. **Sigma notation:** a compact way to write a long sum, and the rules for manipulating one. *This is the part you will use every week.* (Section 2)
2. **Closed forms:** finite sums that collapse to a single formula. (Section 3)
3. **Partial sums**, and what **convergence** means. (Section 4)
4. The **geometric series**, the one worth memorizing, and the answer to the dosing question. (Sections 4 and 5)

Along the way there are many worked examples, most of them drawn from clinical and public-health data, and each section ends with extra practice problems with full solutions.

The notebook for this lecture is `U1-4_Series-1_Convergence.ipynb`. It writes sums as loops, checks closed forms against those loops, plots partial sums closing in on their limits (including the $0.999\dots = 1$ example), and applies the geometric series to a drug’s half-life.

2 Sigma Notation

2.1 Sequence First, Then Series

Two words sound alike here and are easy to mix up, so we pin them down first.

Definition: Sequence

A **sequence** is an ordered list of numbers, $a_1, a_2, a_3, \dots$, one number for each index. It is a function whose domain is the counting numbers: feed it n , it returns a_n .

The small number written below the letter is called a **subscript**, or **index**: a_1 is the first number in the list, a_2 the second, and a_n the n th. Thinking of a sequence as a function is exactly the Unit 1-3 idea: the input is a position n , and the output is the number a_n that sits there.

Definition: Series

A **series** is what you get when you **add up** a sequence:

$$S = a_1 + a_2 + a_3 + \dots + a_n.$$

The distinction is worth keeping straight: a sequence is the *list*, a series is the *total*. In a data table, a column of values is the sequence; the number at the bottom of the column, where the column is added up, is the series.

Sequences come to us in two ways: from a *formula* that says how to compute the n th entry, or from *data*, where the n th entry is simply whatever was recorded in position n . Here are examples of each.

Example 2.1: A sequence from a formula. Take $a_n = n^2 + 1$. To list the sequence, feed in $n = 1, 2, 3, \dots$ one at a time:

$$a_1 = 1^2 + 1 = 2, \quad a_2 = 2^2 + 1 = 5, \quad a_3 = 10, \quad a_4 = 17, \quad a_5 = 26, \quad \dots$$

The series made from the first five terms is the single number

$$S = 2 + 5 + 10 + 17 + 26 = 60.$$

Notice the two different kinds of answer: the sequence is five numbers, and the series is one.

A few more sequences given by formulas, to get used to reading them:

Formula	First few terms	What it describes
$a_n = 2n$	2, 4, 6, 8, ...	the even numbers
$a_n = \frac{1}{n}$	1, $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$, ...	shrinks toward 0
$a_n = (-1)^n$	-1, 1, -1, 1, ...	a sign that flips every step
$a_n = 100(0.5)^{n-1}$	100, 50, 25, 12.5, ...	what is left of each dose in Section 1

The third one is worth remembering: $(-1)^n$ is -1 when n is odd and $+1$ when n is even, so multiplying a term by it is the standard way to make signs alternate.

Example 2.2: A sequence from data. A patient records her systolic blood pressure (in mmHg) every morning for a week:

Day i	1	2	3	4	5	6	7
Reading a_i	128	131	125	134	129	127	136

This is a sequence: $a_3 = 125$ is the reading on day 3, and the order matters, because the readings belong to particular days. No formula produces these numbers; the “rule” is “look up day i .” The series is the weekly total, $a_1 + a_2 + \cdots + a_7 = 910$, and dividing it by 7 gives her average reading for the week, 130 mmHg. Almost every summary statistic starts exactly like this: a sequence from data, turned into a series.

A non-example. The set $\{125, 128, 131\}$ is *not* a sequence. A set (Unit 1-2) has no order and no repeated elements, while a sequence has positions, and the same value can sit in several positions: two days can have the same reading, and they still count as two entries. For the same reason, 1, 2, 3 and 3, 1, 2 are two *different* sequences, even though they contain the same numbers. They do have the same sum, since the order in which you add finitely many numbers never changes the total. A sequence cares about order; a finite series does not.

Watch out: “Series” in pandas

In the Python library pandas, a **Series** is a single *column* of data, which in the vocabulary of these notes is a **sequence**. The mathematical series, the total, is what `.sum()` returns. Same word, two different meanings; the context always tells you which one is meant.

2.2 Writing a Long Sum Compactly

Writing out $a_1 + a_2 + \cdots + a_{100}$ term by term is impractical, and writing out a sum that continues forever is impossible. **Sigma notation** says the same thing in one symbol.

Definition: Sigma notation

$$\sum_{i=1}^n a_i = a_1 + a_2 + a_3 + \cdots + a_n$$

The symbol has four parts, and each one answers a question about the sum:

- $\sum$, the Greek capital letter sigma (the Greek “S,” for “sum”), means “add up what follows.”
- i is the **index**: a counter that takes one value at a time.
- $i = 1$ below the sigma and n above it say where the counter **starts** and where it **stops**. Both endpoints are included.
- a_i is the **general term**: the recipe for what is added, written in terms of i .

Read it out loud as “the sum, as i runs from 1 to n , of a_i .”

Shorthand. When the range is obvious or beside the point, the limits get left off, and you will see $\sum_i a_i$, or even just $\sum a_i$. In statistics, $\sum x_i$ means “add up x_i over all n data points.” The

shorthand saves ink, but it also hides information, so write the limits whenever there is any doubt about where the sum starts and stops.

Reading a sigma: worked examples

The skill to build first is *reading*: given a sigma, write out what it adds up and find the total. The recipe is always the same. Identify the index, the start, the stop, and the general term; then substitute each value of the index in turn.

Example 2.3: The first five odd numbers. Consider

$$\sum_{i=1}^5 (2i - 1).$$

Read aloud, this is “the sum, as i runs from 1 to 5, of $2i - 1$.” The index is i , it starts at 1 and stops at 5, and the general term is $2i - 1$. Substituting each value of i :

$$\underbrace{(2 \cdot 1 - 1)}_{i=1} + \underbrace{(2 \cdot 2 - 1)}_{i=2} + \underbrace{(2 \cdot 3 - 1)}_{i=3} + \underbrace{(2 \cdot 4 - 1)}_{i=4} + \underbrace{(2 \cdot 5 - 1)}_{i=5} = 1 + 3 + 5 + 7 + 9 = 25.$$

So this sigma is the sum of the first five odd numbers, and it comes to 25. (That $25 = 5^2$ is no accident; Section 3 shows why.)

Example 2.4: A sum that does not start at 1. Consider

$$\sum_{k=2}^6 \frac{1}{k}.$$

Two things are different here: the index is called k instead of i (the letter does not matter), and it starts at 2. The terms are $k = 2, 3, 4, 5, 6$, five of them:

$$\sum_{k=2}^6 \frac{1}{k} = \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6}.$$

To add fractions, put them over a common denominator; 60 works for all five. Then

$$\frac{30}{60} + \frac{20}{60} + \frac{15}{60} + \frac{12}{60} + \frac{10}{60} = \frac{87}{60} = \frac{29}{20} = 1.45.$$

Example 2.5: Alternating signs. Consider

$$\sum_{i=1}^4 (-1)^i i.$$

The general term has two pieces: i itself, and the sign switch $(-1)^i$, which is -1 for odd i and $+1$ for even i . So

$$\sum_{i=1}^4 (-1)^i i = (-1)(1) + (+1)(2) + (-1)(3) + (+1)(4) = -1 + 2 - 3 + 4 = 2.$$

If you wanted the signs the other way round, starting with a plus, you would write $(-1)^{i+1}$ instead: it is $+1$ when i is odd.

Example 2.6: A sum over data. Five patients have fasting glucose readings (in mg/dL) of

$$x_1 = 92, \quad x_2 = 105, \quad x_3 = 88, \quad x_4 = 130, \quad x_5 = 110.$$

Then

$$\sum_{i=1}^5 x_i = 92 + 105 + 88 + 130 + 110 = 525.$$

Here the general term x_i is not a formula at all: it means “look up row i .” This is how sigma notation is used most of the time in data science, and it is why the notation is written with a subscript rather than a formula: the recipe for a term can be a lookup. We return to these five readings several times below.

Counting the terms

A sum from $i = m$ to $i = n$, with both ends included, has

$$n - m + 1 \text{ terms.}$$

So $\sum_{i=1}^n$ has n terms, $\sum_{k=2}^6$ has $6 - 2 + 1 = 5$ terms (as we just saw), $\sum_{i=0}^{n-1}$ has n terms, and $\sum_{i=3}^{10}$ has 8 terms, not 7. The “ $+1$ ” is the same one that shows up when you count fence posts: a fence from post 3 to post 10 spans 7 gaps but has 8 posts, and a sum counts posts. You will need this count every time a constant term is summed (Section 2.5) and every time the geometric formula is used (Section 3).

Watch out: Parentheses matter

By convention, a sigma applies only to the term written right after it. So $\sum_{i=1}^3 i + 1$ is usually read as $\left(\sum_{i=1}^3 i\right) + 1 = 6 + 1 = 7$, while

$$\sum_{i=1}^3 (i + 1) = 2 + 3 + 4 = 9.$$

Whenever the general term has more than one piece, put parentheses around the whole term.

Going the other way: from a written-out sum to sigma notation

The reverse skill, *writing* a sum in sigma notation, is what you need when you want to hand a long sum to a formula or to a computer. It is a matter of spotting the pattern.

Procedure: writing a sum in sigma notation

1. Look at what changes from one term to the next, and what stays the same.
2. Write the general term as a formula in the position i .
3. Choose the start and the stop so that the index runs over exactly the terms you have.
4. Check: substitute the start and the stop into your general term, and confirm that you get the first and the last terms.

Example 2.7: Squares. Write $1 + 4 + 9 + \cdots + 100$ in sigma notation. The terms are $1^2, 2^2, 3^2, \dots$, so the general term is i^2 . The last term is $100 = 10^2$, so the index stops at 10:

$$1 + 4 + 9 + \cdots + 100 = \sum_{i=1}^{10} i^2.$$

Check: $i = 1$ gives 1 and $i = 10$ gives 100.

Example 2.8: Evenly spaced terms. Write $5 + 9 + 13 + 17 + 21$ in sigma notation. Each term is 4 more than the one before, so the general term should contain $4i$: stepping i up by 1 then raises the term by 4. With $i = 1$ we need 5, and $4 \cdot 1 = 4$ falls short by 1, so the term is $4i + 1$. There are five terms:

$$5 + 9 + 13 + 17 + 21 = \sum_{i=1}^5 (4i + 1).$$

Check the last term: $4 \cdot 5 + 1 = 21$. The same reasoning works for any list that goes up by a fixed step d : the general term is $d \cdot i$ plus whatever constant makes the first term come out right.

Example 2.9: Halving. Write $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16}$ in sigma notation. Each denominator is a power of 2: $2^1, 2^2, 2^3, 2^4$. So

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} = \sum_{i=1}^4 \frac{1}{2^i}.$$

Check: $i = 4$ gives $\frac{1}{16}$.

Example 2.10: Alternating and shrinking. Write $1 - \frac{1}{3} + \frac{1}{9} - \frac{1}{27}$ in sigma notation. Here each term is the previous one multiplied by $-\frac{1}{3}$: that one factor both shrinks the size and flips the sign. So each term is a power of $-\frac{1}{3}$, and starting the index at 0 makes the first term $\left(-\frac{1}{3}\right)^0 = 1$:

$$1 - \frac{1}{3} + \frac{1}{9} - \frac{1}{27} = \sum_{i=0}^3 \left(-\frac{1}{3}\right)^i.$$

Check the last term: $\left(-\frac{1}{3}\right)^3 = -\frac{1}{27}$. Starting at 0 here is a choice that keeps the formula clean; starting at 1 would also work, with the term written $\left(-\frac{1}{3}\right)^{i-1}$ and the stop moved to 4 (the next subsections explain this kind of shift).

2.3 Sigma Is a For-Loop

The notation is exactly the loop you would write in code: start a counter, run it up to the limit, and accumulate a running total. Here is the same sum written both ways.

$$\begin{aligned}\sum_{i=1}^4 i^2 &= 1^2 + 2^2 + 3^2 + 4^2 \\ &= 1 + 4 + 9 + 16 \\ &= 30\end{aligned}$$

```
total = 0
for i in range(1, 5):
    total += i**2
```

Each piece of the sigma has a partner in the loop:

In the sigma	In the loop
the sum starts from nothing	<code>total = 0</code>
the index i starts at 1 and stops at 4	<code>for i in range(1, 5):</code>
the general term i^2 is added each time	<code>total += i**2</code>

It is worth tracing the loop by hand once, pass by pass, to see the running total build up:

Pass	Value of i	Term added, i^2	total after this pass
1	1	1	1
2	2	4	5
3	3	9	14
4	4	16	30

The last column, the running total after each pass, will matter again in Section 4, where these running totals get a name (*partial sums*) and become the key to adding infinitely many numbers.

Notice the stopping value. The upper limit is *inclusive* in mathematics, so $\sum_{i=1}^4$ includes $i = 4$. Python's `range` is *exclusive* at the top, so `range(1, 5)` produces 1, 2, 3, 4 and stops *before* 5. That mismatch costs everyone a bug at least once.

Watch out: Translating limits into Python

To loop over $\sum_{i=m}^n$, write `range(m, n + 1)`. Going the other way, a loop over `range(m, k)` computes a sum that stops at $i = k - 1$. A second off-by-one trap: Python counts positions in a list from 0, so the mathematical x_1 is `x[0]` in code, and x_n is `x[n - 1]`.

Expanding a sigma: the mechanical procedure

To expand a sigma, substitute the starting index into the general term, then the next value, and so on up to the upper limit, joining the results with plus signs.

Example 2.11: From sigma to loop. Translate $\sum_{i=3}^7 (2i + 1)$ into a loop, and find its value. The index runs from 3 to 7 inclusive, so the loop is `for i in range(3, 8):` with `total += 2*i`

+ 1 inside. By hand, the terms are

$$\underbrace{7}_{i=3} + \underbrace{9}_{i=4} + \underbrace{11}_{i=5} + \underbrace{13}_{i=6} + \underbrace{15}_{i=7} = 55.$$

There are $7 - 3 + 1 = 5$ terms, matching the five passes of the loop.

Example 2.12: From loop to sigma. What sum does this code compute?

```
total = 0
for j in range(0, 10):
    total += 3*j
```

The counter is `j`, it starts at 0, and because `range` stops before 10, the last value is 9. The term is $3j$. So the code computes

$$\sum_{j=0}^9 3j = 0 + 3 + 6 + 9 + \cdots + 27 = 135.$$

The common mistake is to write the top limit as 10, which would add one more term, 30, and give 165.

Example 2.13: Letting the library run the loop. When the numbers are stored in an array, you rarely write the loop yourself. For the five glucose readings,

```
import numpy as np
x = np.array([92, 105, 88, 130, 110])
x.sum()          # 525, the same as np.sum(x)
```

The call `x.sum()` is $\sum_{i=1}^5 x_i$: it runs the loop for you (and much faster than a hand-written loop on a large array). Reading a sigma and reading a line like this are the same skill.

2.4 The Index Is a Dummy Variable

The index has no meaning outside the sum. Its only job is to run through the values from the start to the stop, so you can rename it and nothing changes:

$$\sum_{i=1}^n a_i = \sum_{j=1}^n a_j = \sum_{k=1}^n a_k.$$

All three say “add up a_1 through a_n .” A variable whose name can be swapped out like this without changing the meaning is called a **dummy variable**. It also means the index never escapes the sum: once the sum is finished, i has no value, and writing i anywhere outside it is meaningless.

Watch out: The index cannot leak out of the sum

A common slip is to write $\sum_{i=1}^n x_i = nx_i$. The right-hand side has an i with no value (which x_i is meant?), so the equation does not say anything. The correct statement uses the mean $\bar{x}$, a single number: $\sum_{i=1}^n x_i = n\bar{x}$ (see Section 2.8). A good habit: after simplifying a sum, check that no index letter is left outside a sigma.

The starting value is a choice, not a law. These two write the same sum:

$$\underbrace{\sum_{i=1}^n 2^{i-1}}_{\text{start at 1}} = 1 + 2 + 4 + \cdots + 2^{n-1} = \underbrace{\sum_{i=0}^{n-1} 2^i}_{\text{start at 0}}$$

Check the first and last terms. On the left, $i = 1$ gives $2^0 = 1$ and $i = n$ gives 2^{n-1} . On the right, $i = 0$ gives $2^0 = 1$ and $i = n - 1$ gives 2^{n-1} . Same first term, same last term, same terms in between.

Shifting the start by one means shifting every appearance of i in the term (and the stopping value) to compensate. Starting at $i = 0$ is common for geometric series, which we meet in Section 3, because then the exponent *is* the index.

Shifting an index is a mechanical procedure, and it is worth doing carefully, because it comes up every time a formula is written for a sum that starts at 0 and your sum starts at 1 (or the other way round).

Procedure: shifting the index

To rewrite $\sum_{i=m}^n a_i$ so that it starts somewhere else:

1. Introduce a new index j and decide how the two are related, say $i = j + s$ for a fixed shift s .
2. Replace every i in the general term by $j + s$.
3. Move both limits by the same amount: j runs from $m - s$ to $n - s$.
4. Check that the first and the last terms are unchanged. (Afterwards you may rename j back to i ; it is a dummy.)

Example 2.14: Starting at 1. Rewrite $\sum_{i=3}^7 i^2$ so that it starts at 1. We want the new index to be 1 when the old one is 3, so $i = j + 2$. The term i^2 becomes $(j + 2)^2$, and the limits 3 and 7 become 1 and 5:

$$\sum_{i=3}^7 i^2 = \sum_{j=1}^5 (j + 2)^2.$$

Check by expanding both: the left is $9 + 16 + 25 + 36 + 49$, and the right is $(1 + 2)^2 + (2 + 2)^2 + \cdots + (5 + 2)^2 = 9 + 16 + 25 + 36 + 49$. Both total 135.

Example 2.15: Starting at 0. Rewrite $\sum_{i=1}^n 3^i$ so that it starts at 0. Now $i = j + 1$, the term 3^i becomes 3^{j+1} , and the limits 1 and n become 0 and $n - 1$:

$$\sum_{i=1}^n 3^i = \sum_{j=0}^{n-1} 3^{j+1}.$$

Check with $n = 4$: the left is $3 + 9 + 27 + 81 = 120$, and the right is $3^1 + 3^2 + 3^3 + 3^4$, the same four terms.

Example 2.16: The dosing sum, both ways. The amount in the body after 10 doses in the opening problem can be written with the index starting at 1 (dose number) or at 0 (number of halvings):

$$\sum_{i=1}^{10} 100(0.5)^{i-1} = \sum_{i=0}^9 100(0.5)^i.$$

Here the shift is $i_{\text{old}} = i_{\text{new}} + 1$, which turns the exponent $i - 1$ into plain i , and moves the limits from 1 and 10 to 0 and 9. Both versions appear in practice. The first matches how a clinician counts (dose 1, dose 2, ...); the second is the form the geometric formula of Section 3 is written for.

Watch out: Shift the term and the limits together

Changing the limits without changing the term changes the sum. For example $\sum_{i=1}^4 i^2 = 1 + 4 + 9 + 16 = 30$, but $\sum_{i=0}^3 i^2 = 0 + 1 + 4 + 9 = 14$. Moving the start from 1 to 0 is only legal if every i in the term becomes $i + 1$ at the same time.

2.5 Rules for Manipulating Sums

Three rules do almost all the work of simplifying a sum. None of them is new: each is just regrouping an ordinary addition.

Three rules for sums

1. Constants pull out:

$$\sum_{i=1}^n c a_i = c \sum_{i=1}^n a_i$$

2. Sums split:

$$\sum_{i=1}^n (a_i + b_i) = \sum_{i=1}^n a_i + \sum_{i=1}^n b_i$$

3. A constant term counts itself:

$$\sum_{i=1}^n c = \underbrace{c + c + \cdots + c}_{n \text{ times}} = nc$$

To see why each one holds, write the sum out in full.

- **Constants pull out** because every term carries the same factor c , and a common factor can be pulled out of an addition:

$$ca_1 + ca_2 + \cdots + ca_n = c(a_1 + a_2 + \cdots + a_n).$$

- **Sums split** because the order and grouping of an addition do not matter. Written out, the left side is $(a_1 + b_1) + (a_2 + b_2) + \cdots + (a_n + b_n)$; gather all the a 's together and all the b 's together and you have the right side.

- **A constant term counts itself** because a term with no i in it does not change from one pass of the loop to the next. The loop runs n times and adds the same c every time, so the total is n copies of c . (It is a common slip to write $\sum_{i=1}^n c = c$, forgetting that the c is added once per pass.)

Here “constant” means *constant with respect to the index*: anything that does not change as i runs, such as a fixed number, a unit conversion factor, or a summary statistic like the mean. The rules are what let you compute with totals you already have, instead of going back to the raw data.

Example 2.17: Using a total you already know. Ten lab values $x_1, \dots, x_{10}$ are known to add up to 40. Find $\sum_{i=1}^{10} (3x_i - 2)$ without knowing the individual values. Split the sum, pull the 3 out of the first piece, and count the constant in the second:

$$\sum_{i=1}^{10} (3x_i - 2) = 3 \sum_{i=1}^{10} x_i - \sum_{i=1}^{10} 2 = 3(40) - 10(2) = 120 - 20 = 100.$$

(Subtracting is adding a negative, so “sums split” covers differences too.) The point is that the ten individual values were never needed: the rules turned a question about ten numbers into a question about one total.

Example 2.18: Converting a column’s units. Four patients have temperatures, in degrees Celsius, of

$$37.0, \quad 38.5, \quad 36.8, \quad 39.1,$$

which add up to $\sum c_i = 151.4$. To convert Celsius to Fahrenheit, use $f_i = 1.8c_i + 32$. The total in Fahrenheit is

$$\sum_{i=1}^4 (1.8c_i + 32) = 1.8 \sum_{i=1}^4 c_i + 4(32) = 1.8(151.4) + 128 = 272.52 + 128 = 400.52.$$

Check it the long way: the four Fahrenheit values are 98.6, 101.3, 98.24 and 102.38, which add to 400.52. Dividing by 4, the mean temperature is 100.13°F, which is exactly $1.8(37.85) + 32$, where 37.85°C is the mean in Celsius. So a linear change of units passes straight through a total and through a mean: convert the summary, not every row. The tempting slip here is to add the 32 only once, which gives $1.8(151.4) + 32 = 304.52$; the rules say 32 is added once *per patient*.

Example 2.19: Deviations from the mean add up to zero. This one is a little more abstract, and it is a fact statistics uses constantly. The mean of n values is $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ (Section 2.8); multiplying both sides by n gives $\sum_{i=1}^n x_i = n\bar{x}$. The *deviation* of value i from the mean is $x_i - \bar{x}$. Add up all the deviations:

$$\sum_{i=1}^n (x_i - \bar{x}) = \sum_{i=1}^n x_i - \sum_{i=1}^n \bar{x} = n\bar{x} - n\bar{x} = 0.$$

The key step is the middle one: $\bar{x}$ is one fixed number (it does not change with i), so it is a constant term and counts itself, giving $n\bar{x}$. Check it on the glucose readings, whose mean is $525/5 = 105$: the deviations are

$$92 - 105 = -13, \quad 0, \quad -17, \quad 25, \quad 5, \quad \text{and} \quad -13 + 0 - 17 + 25 + 5 = 0.$$

In words, the mean is the balance point of the data: the values above it and the values below it exactly cancel. This is also why Unit 4, when it measures how spread out data are, squares the deviations before adding them; the plain deviations always add to zero, whatever the data.

Watch out: Two slips with the rules

- **A constant term counts the terms, not the top limit.** $\sum_{i=5}^{12} 4$ has $12 - 5 + 1 = 8$ terms, so it equals $8 \cdot 4 = 32$, not $12 \cdot 4 = 48$.
- **Only constants pull out.** Something that changes with i must stay inside: $\sum_{i=1}^n i x_i$ is *not* $i \sum_{i=1}^n x_i$. The version on the right has let the index leak out of the sum.

2.6 A Warning About What Does *Not* Split

The first two rules are together called **linearity**, and linearity is about addition only. Products and powers do **not** pass through a sum:

$$\sum_{i=1}^n a_i b_i \neq \left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right) \quad \sum_{i=1}^n a_i^2 \neq \left(\sum_{i=1}^n a_i \right)^2$$

When a rule seems plausible, the fastest way to test it is with the smallest possible example. Take $n = 2$ with $a = (1, 1)$ and $b = (1, 1)$, that is, $a_1 = a_2 = 1$ and $b_1 = b_2 = 1$. Then

$$\sum_{i=1}^2 a_i b_i = 1 + 1 = 2, \quad \text{while} \quad \left(\sum_{i=1}^2 a_i \right) \left(\sum_{i=1}^2 b_i \right) = 2 \cdot 2 = 4.$$

The same example settles the second inequality, since b is the same list as a : $\sum a_i^2 = 1 + 1 = 2$, while $(\sum a_i)^2 = 2^2 = 4$.

The reason is visible if you multiply out the right-hand side for $n = 2$:

$$(a_1 + a_2)(b_1 + b_2) = a_1 b_1 + a_2 b_2 + \underbrace{a_1 b_2 + a_2 b_1}_{\text{extra "cross" terms}}.$$

The product of the sums contains every term of $\sum a_i b_i$ *plus* cross terms that pair an entry of one list with a different entry of the other. Those extra terms are the gap.

Example 2.20: A slightly bigger check. Take $n = 3$ with $a = (1, 2, 3)$ and $b = (4, 5, 6)$. Then

$$\sum_{i=1}^3 a_i b_i = 1 \cdot 4 + 2 \cdot 5 + 3 \cdot 6 = 4 + 10 + 18 = 32, \quad \left(\sum_{i=1}^3 a_i \right) \left(\sum_{i=1}^3 b_i \right) = 6 \cdot 15 = 90.$$

Multiplying out $(a_1 + a_2 + a_3)(b_1 + b_2 + b_3)$ gives $3 \times 3 = 9$ products, one for every pairing of an a with a b , while $\sum a_i b_i$ keeps only the 3 matched pairs. The other 6 products, the cross terms, add up to $90 - 32 = 58$. For squares, $\sum a_i^2 = 1 + 4 + 9 = 14$, while $(\sum a_i)^2 = 6^2 = 36$. As n grows, the number of cross terms grows much faster than the number of matched terms, so the gap only gets worse.

Example 2.21: Rates in two clinics. Division does not pass through a sum either, and this one causes real mistakes in reports. Two clinics run a screening test:

	Positive results a_i	Tests run b_i	Positivity rate a_i/b_i
Clinic 1	2	10	0.2
Clinic 2	45	90	0.5
Total	47	100	

The sum of the two rates is $\sum a_i/b_i = 0.2 + 0.5 = 0.7$, while the ratio of the totals is $\sum a_i/\sum b_i = 47/100 = 0.47$: not equal. The practical version of this is about averages. The average of the two clinics' rates is $0.7/2 = 0.35$, but the positivity rate across all 100 tests is 0.47. Neither is “wrong”; they answer different questions (“what is a typical clinic’s rate?” versus “what fraction of all tests were positive?”). They differ because clinic 2 ran nine times as many tests, and the pooled rate weights it accordingly. Averaging rates across groups of different sizes, and reporting the result as an overall rate, is a classic data-analysis error.

One more quick check, for square roots: with the two values 9 and 16, $\sqrt{9} + \sqrt{16} = 3 + 4 = 7$, but $\sqrt{9 + 16} = \sqrt{25} = 5$.

Example 2.22: What you can do instead: expand first. A square does not pass through a sum, but you can still simplify $\sum (x_i + 1)^2$. The trick is to expand each term *first*, using ordinary algebra, and only then use linearity:

$$\sum_{i=1}^n (x_i + 1)^2 = \sum_{i=1}^n (x_i^2 + 2x_i + 1) = \sum_{i=1}^n x_i^2 + 2 \sum_{i=1}^n x_i + n.$$

Check with $x = (1, 2, 3)$: directly, $(1 + 1)^2 + (2 + 1)^2 + (3 + 1)^2 = 4 + 9 + 16 = 29$; by the formula, $14 + 2(6) + 3 = 29$. So a sum involving squares can be computed from two totals, $\sum x_i$ and $\sum x_i^2$, but both are needed; neither can be recovered from the other.

That gap is not a nuisance. It is the heart of the variance formula in Unit 4, where $\sum x_i^2$ and $(\sum x_i)^2$ appear side by side (the second one divided by n), and the difference between them measures the spread of the data.

You can see this already with two tiny data sets that have the same total, 15, and the same mean, 5:

Data	$\sum x_i^2$	$(\sum x_i)^2/n$	Gap
5, 5, 5 (no spread at all)	$25 + 25 + 25 = 75$	$15^2/3 = 75$	0
1, 5, 9 (spread out)	$1 + 25 + 81 = 107$	$15^2/3 = 75$	32

When every value is the same, the gap is exactly zero; as soon as the values spread out, $\sum x_i^2$ pulls ahead of $(\sum x_i)^2/n$. Unit 4 turns this gap into the variance.

2.7 Practice: Reading Sigma Notation

Exercise 2.1

1. Write out every term of $\sum_{i=1}^3 \frac{a_i}{b_i}$.
2. Write out every term of $\sum_{i=0}^5 \frac{i^2}{i+1}$, and find the total.
3. Rewrite $3 + 6 + 9 + 12 + 15$ in sigma notation.
4. Simplify $\sum_{i=1}^n (4a_i + 7)$ using the rules of Section 2.5.

Solution to Exercise 2.1

1. Substitute $i = 1$, then $i = 2$, then $i = 3$ into the general term. The index rides along on *both* letters:

$$\sum_{i=1}^3 \frac{a_i}{b_i} = \frac{a_1}{b_1} + \frac{a_2}{b_2} + \frac{a_3}{b_3}.$$

This is a sum of three separate fractions. It is *not* the single fraction you would get by adding up all the tops and all the bottoms,

$$\frac{a_1 + a_2 + a_3}{b_1 + b_2 + b_3}.$$

2. The index starts at $i = 0$, which is legal; that term is simply zero. Substituting each value of i from 0 to 5:

i	i^2	$i + 1$	term $i^2/(i + 1)$	over 60
0	0	1	0	0/60
1	1	2	1/2	30/60
2	4	3	4/3	80/60
3	9	4	9/4	135/60
4	16	5	16/5	192/60
5	25	6	25/6	250/60

To add fractions, put them over a common denominator; the smallest one that every denominator 1, 2, 3, 4, 5, 6 divides into is 60. Then

$$\begin{aligned}\sum_{i=0}^5 \frac{i^2}{i+1} &= \frac{0}{1} + \frac{1}{2} + \frac{4}{3} + \frac{9}{4} + \frac{16}{5} + \frac{25}{6} \\ &= \frac{0 + 30 + 80 + 135 + 192 + 250}{60} = \frac{687}{60} = 11.45.\end{aligned}$$

3. Each term is 3 times its position in the list ($3 = 3 \cdot 1$, $6 = 3 \cdot 2$, up to $15 = 3 \cdot 5$), and there are five terms, so

$$3 + 6 + 9 + 12 + 15 = \sum_{i=1}^5 3i.$$

Starting the index somewhere else is equally correct, as long as the term shifts to compensate. Starting at $i = 0$ means replacing i by $i + 1$ in the term, $3(i + 1) = 3i + 3$, and stopping one sooner: $\sum_{i=0}^4 (3i + 3)$. Check it: $i = 0$ gives 3 and $i = 4$ gives 15.

4. Apply the three rules one at a time. Split the sum, pull the constant 4 out of the first piece, and count the sevens in the second:

$$\sum_{i=1}^n (4a_i + 7) = \sum_{i=1}^n 4a_i + \sum_{i=1}^n 7 = 4 \sum_{i=1}^n a_i + 7n.$$

In words: adding up $4a_i + 7$ over n rows is the same as four times the total of the a_i , plus one 7 for each of the n rows.

A few more problems, for practice with reading, writing, shifting and simplifying.

Exercise 2.2

1. Expand $\sum_{k=2}^5 (k^2 - k)$ and find its value.
2. How many terms does $\sum_{i=4}^{15} a_i$ have?
3. Write $\frac{1}{2} + \frac{2}{3} + \frac{3}{4} + \cdots + \frac{9}{10}$ in sigma notation.
4. Write $2 - 4 + 8 - 16 + 32$ in sigma notation, and find its value.

Solution to Exercise 2.2

1. The index k runs over 2, 3, 4, 5. Substituting each:

$$\underbrace{(4 - 2)}_{k=2} + \underbrace{(9 - 3)}_{k=3} + \underbrace{(16 - 4)}_{k=4} + \underbrace{(25 - 5)}_{k=5} = 2 + 6 + 12 + 20 = 40.$$

2. From 4 to 15 inclusive: $15 - 4 + 1 = 12$ terms. (Answering 11 is the fence-post mistake.)
3. The top of each fraction counts 1, 2, 3, ... and the bottom is always one more than the top. So the general term is $\frac{i}{i+1}$, and the last term $\frac{9}{10}$ has $i = 9$:

$$\frac{1}{2} + \frac{2}{3} + \cdots + \frac{9}{10} = \sum_{i=1}^9 \frac{i}{i+1}.$$

Check: $i = 1$ gives $\frac{1}{2}$ and $i = 9$ gives $\frac{9}{10}$.

4. The sizes are $2^1, 2^2, 2^3, 2^4, 2^5$, and the signs alternate starting with a plus. A plus on the odd positions calls for $(-1)^{i+1}$:

$$2 - 4 + 8 - 16 + 32 = \sum_{i=1}^5 (-1)^{i+1} 2^i.$$

Check: $i = 2$ gives $(-1)^3 \cdot 4 = -4$, as it should. The value is $2 - 4 + 8 - 16 + 32 = 22$.

Exercise 2.3

1. Rewrite $\sum_{i=4}^9 (i-3)^2$ so that its index starts at 1, and find its value.
2. What sum does this code compute? Write it in sigma notation and find its value.

```
total = 0
for k in range(2, 8):
    total += k*(k - 1)
```

Solution to Exercise 2.3

1. We want the new index j to be 1 when $i = 4$, so $i = j + 3$. Then $(i-3)^2$ becomes $(j+3-3)^2 = j^2$, and the limits 4 and 9 become 1 and 6:

$$\sum_{i=4}^9 (i-3)^2 = \sum_{j=1}^6 j^2 = 1 + 4 + 9 + 16 + 25 + 36 = 91.$$

The shift made the sum much easier to read: it is just the first six squares. Check against the original: $i = 4$ gives $(4-3)^2 = 1$ and $i = 9$ gives $6^2 = 36$.

2. The counter k starts at 2, and `range(2, 8)` stops before 8, so the last value is 7. The term is $k(k-1)$:

$$\sum_{k=2}^7 k(k-1) = 2 + 6 + 12 + 20 + 30 + 42 = 112.$$

That is $7 - 2 + 1 = 6$ terms, one for each pass of the loop.

Exercise 2.4

A column of $n = 20$ lab values has $\sum_{i=1}^{20} x_i = 150$ and $\sum_{i=1}^{20} x_i^2 = 1400$. Without knowing the individual values, find:

1. $\sum_{i=1}^{20} (2x_i - 3)$, and the mean of the new column $2x_i - 3$;
2. $\sum_{i=1}^{20} (x_i + 1)^2$.
3. A classmate computes part 2 as $(\sum x_i)^2 + \dots$. Explain why $(\sum x_i)^2 = 22,500$ is of no use here.

Solution to Exercise 2.4

1. Split, pull out the 2, count the 3's:

$$\sum_{i=1}^{20} (2x_i - 3) = 2(150) - 20(3) = 300 - 60 = 240.$$

The mean of the new column is $240/20 = 12$. As a check, the mean of the original column is $150/20 = 7.5$, and $2(7.5) - 3 = 12$: the same linear change passes straight through the mean, as in the temperature example.

2. Expand the square first, then split:

$$\sum_{i=1}^{20} (x_i + 1)^2 = \sum x_i^2 + 2 \sum x_i + \sum 1 = 1400 + 2(150) + 20 = 1720.$$

3. The sum in part 2 is built from the squares of the *individual* values, x_i^2 , one per row. The number $(\sum x_i)^2$ squares the *total*, which mixes every row with every other row through the cross terms. Since a square does not pass through a sum, $(\sum x_i)^2$ and $\sum x_i^2$ are different quantities (22,500 versus 1400 here), and only the second one appears when you expand $(x_i + 1)^2$.

2.8 Sums You Already Use

Every one of these is sigma notation wearing a different name, and each returns later in the course:

- **Mean** of n values (Unit 4): add up all the values and divide by how many there are.

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

- **Sum of squared errors** (Unit 3, optimization): y_i is the true value in row i and $\hat{y}_i$ (read “ y -hat”) is the model’s prediction for it. Square each error so that misses in either direction

count against the model, then add.

$$E = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- **Dot product** of two lists (Unit 2, vectors): multiply matching entries and add.

$$\mathbf{v} \cdot \mathbf{w} = \sum_{i=1}^p v_i w_i$$

- **Linear model** with p features (everywhere): each feature x_j gets a weight w_j , the weighted features are added up, and a constant b is added on.

$$\hat{y} = b + \sum_{j=1}^p w_j x_j$$

Notice that the last one uses j as its index; by the dummy-variable rule, that changes nothing. Once you can read a sigma you can read nearly every definition in machine learning: the notation is how the field writes its ideas down.

Here is each one computed on small, concrete numbers, so the formulas have something to hold on to.

Example 2.23: A mean. For the five fasting glucose readings 92, 105, 88, 130, 110 (mg/dL),

$$\bar{x} = \frac{1}{5} \sum_{i=1}^5 x_i = \frac{525}{5} = 105 \text{ mg/dL.}$$

Rearranged, the definition says $\sum x_i = n\bar{x}$: a total is the mean times the count. That rearrangement is the one used in the “deviations add to zero” example of Section 2.5. (The bar in $\bar{x}$ means “average.” In the Unit 2 slides a bar over a letter is used for a different purpose, to mark a vector; the context always tells you which.)

Example 2.24: A count is a sum too. How many of the five patients have a fasting glucose at or above 126 mg/dL, the usual cutoff for diabetes? Define an **indicator** for each patient: $d_i = 1$ if $x_i \geq 126$ and $d_i = 0$ otherwise. For our data the indicators are 0, 0, 0, 1, 0, and

$$\sum_{i=1}^5 d_i = 0 + 0 + 0 + 1 + 0 = 1 \text{ patient,} \quad \frac{1}{5} \sum_{i=1}^5 d_i = 0.2.$$

So counting is summing 1’s, and the *mean* of an indicator is a *proportion*: 20% of these patients are at or above the cutoff. This is exactly what the pandas idiom `(x >= 126).sum()` does (and `(x >= 126).mean()` for the proportion): the comparison produces a column of **True/False**, which Python treats as 1/0, and the sum counts them.

Example 2.25: A sum of squared errors. A model predicts systolic blood pressure for four patients:

Patient i	Actual y_i	Predicted $\hat{y}_i$	Error $y_i - \hat{y}_i$	Squared error
1	120	118	2	4
2	135	138	-3	9
3	128	128	0	0
4	142	137	5	25

So $E = \sum_{i=1}^4 (y_i - \hat{y}_i)^2 = 4 + 9 + 0 + 25 = 38$. Notice why the errors are squared: the raw errors add to $2 - 3 + 0 + 5 = 4$, because the model's miss on patient 2 (too high) partly cancels its misses on patients 1 and 4 (too low). Squaring makes every miss count against the model. A training algorithm's whole job is to choose predictions that make E as small as possible.

Example 2.26: A dot product. A clinic orders 3 boxes of gloves at \$4.00 each, 10 packs of gauze at \$1.50 each, and 2 thermometers at \$12.00 each. With quantities $q = (3, 10, 2)$ and prices $p = (4, 1.5, 12)$, the bill is

$$\sum_{i=1}^3 q_i p_i = 3(4) + 10(1.5) + 2(12) = 12 + 15 + 24 = 51 \text{ dollars.}$$

That is a dot product: multiply matching entries, then add. Note that it is *not* $(\sum q_i)(\sum p_i) = 15 \times 17.5 = 262.5$, which would charge every item at every price.

Example 2.27: A linear model. Suppose (with weights made up for illustration, not a real clinical model) that a model predicts systolic blood pressure from $p = 3$ features: $x_1 =$ age in years, $x_2 =$ body mass index, and $x_3 = 1$ for a smoker and 0 otherwise, with

$$\hat{y} = 90 + 0.5x_1 + 1.2x_2 + 8x_3 = b + \sum_{j=1}^3 w_j x_j, \quad b = 90, \quad w = (0.5, 1.2, 8).$$

For a 50-year-old smoker with a body mass index of 30,

$$\hat{y} = 90 + 0.5(50) + 1.2(30) + 8(1) = 90 + 25 + 36 + 8 = 159 \text{ mmHg.}$$

The sigma part, $\sum w_j x_j = 69$, is itself a dot product, of the weights with the features. Real models have hundreds or thousands of features, and sigma notation is what lets them be written in one line.

2.9 The Other Big Operator: $\prod$

The same idea with multiplication instead of addition uses a capital pi, the Greek “P,” for “product”:

$$\prod_{i=1}^n a_i = a_1 \cdot a_2 \cdot a_3 \cdots a_n \quad \text{for instance} \quad \prod_{i=1}^n i = n!$$

Everything about reading a sigma carries over: the index, the start and stop, and the general term all work the same way. The only difference is that the terms are multiplied rather than added. The example on the right multiplies $1 \cdot 2 \cdot 3 \cdots n$, which is the number called n **factorial**, written $n!$.

Example 2.28: Reading products. Expanding a product works exactly like expanding a sum, with times signs in place of plus signs:

$$\prod_{i=1}^5 i = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 = 120 = 5!,$$

$$\prod_{i=1}^3 (i+1) = 2 \cdot 3 \cdot 4 = 24,$$

$$\prod_{i=1}^3 0.5 = 0.5 \cdot 0.5 \cdot 0.5 = 0.125.$$

The last line is the product version of “a constant term counts itself”: a constant multiplied by itself n times is c^n , not nc . It also has a meaning from the opening problem: after three half-lives, $0.5^3 = 0.125$, or one eighth, of a dose is left.

Logarithms turn products into sums

For positive numbers $a_1, \dots, a_n$,

$$\ln \left(\prod_{i=1}^n a_i \right) = \sum_{i=1}^n \ln a_i.$$

This is the logarithm rule from Unit 1-3, $\ln(ab) = \ln a + \ln b$, applied over and over: the log of a product of n numbers is the sum of their n logs. (The logarithm is only defined for positive inputs, which is why the numbers must be positive.)

Example 2.29: Growth factors. A bacterial culture in a lab grows by a factor of 1.2 on day 1, 1.5 on day 2, and 0.9 on day 3 (it shrank a little that day). The overall growth factor is the product,

$$\prod_{i=1}^3 g_i = 1.2 \times 1.5 \times 0.9 = 1.62,$$

so the culture ends 62% larger than it started. On the log scale, the same calculation is a sum:

$$\ln 1.2 + \ln 1.5 + \ln 0.9 \approx 0.1823 + 0.4055 + (-0.1054) = 0.4824 \approx \ln 1.62.$$

Growth factors multiply, but their logs add. The shrinking day shows up as a negative log, since $\ln$ of a number below 1 is negative.

This is not a curiosity. In statistics, the probability of a whole data set is often a product of one probability per row. Each probability is less than 1, and multiplying thousands of them together collapses to zero on a computer, because the result becomes smaller than the smallest number the machine can store (Unit 1-5). So statistics works with the **log-likelihood**, the $\sum \ln$ on the right, instead of the product on the left: a sum of thousands of logs is a perfectly ordinary number. The logarithm rules from Unit 1-3 are what make the swap legal.

Example 2.30: A likelihood, two ways. A model assigns probabilities 0.9, 0.8, 0.95, 0.7 and 0.85 to what actually happened to five patients. The *likelihood* is their product, and the

log-likelihood is the sum of their logs:

$$\prod_{i=1}^5 p_i = 0.9 \times 0.8 \times 0.95 \times 0.7 \times 0.85 \approx 0.4070,$$

$$\sum_{i=1}^5 \ln p_i \approx -0.1054 - 0.2231 - 0.0513 - 0.3567 - 0.1625 = -0.8990 \approx \ln 0.4070.$$

With five rows, both are fine. Now imagine 2000 rows, each with probability 0.5. The true likelihood is $0.5^{2000} = 2^{-2000}$, a number with about 600 zeros after the decimal point. The smallest positive number a standard (double-precision) computer number can hold is about 5×10^{-324} , so multiplying the 2000 factors one at a time in Python returns exactly 0.0: every row's information is lost. The log-likelihood, $\sum \ln p_i = 2000 \ln 0.5 \approx -1386.29$, is an ordinary number that the computer handles with no trouble. Unit 1-5 explains where that storage limit comes from.

Watch out: What logarithms and products do not do

- **The log of a sum is not the sum of the logs.** The rule is for products only: $\ln(1 + 1) = \ln 2 \approx 0.693$, but $\ln 1 + \ln 1 = 0$.
- **A constant comes out of a product as a power.** $\prod_{i=1}^n c a_i = c^n \prod_{i=1}^n a_i$, because the c appears once in each of the n factors. For instance $\prod_{i=1}^3 2a_i = (2a_1)(2a_2)(2a_3) = 8 a_1 a_2 a_3$, not $2 a_1 a_2 a_3$.

Exercise 2.5

1. Evaluate $\prod_{i=1}^4 2i$, and check your answer using the watch-out above.
2. A cell culture grows by factors 1.1, 0.8 and 1.25 over three days. Find the overall growth factor, and the sum of the three logs.
3. True or false: $\ln(\sum_{i=1}^n a_i) = \sum_{i=1}^n \ln a_i$ for positive a_i . Justify with the smallest example you can.

Solution to Exercise 2.5

1. Expanding, $\prod_{i=1}^4 2i = 2 \cdot 4 \cdot 6 \cdot 8 = 384$. The check: each factor is $2 \cdot i$, so the constant 2 comes out as $2^4 = 16$, leaving $\prod_{i=1}^4 i = 4! = 24$; and $16 \times 24 = 384$.
2. The overall factor is $1.1 \times 0.8 \times 1.25 = 1.1$: after a bad day and a good day that exactly undo each other ($0.8 \times 1.25 = 1$), the culture is 10% larger. The logs are $\ln 1.1 \approx 0.0953$, $\ln 0.8 \approx -0.2231$ and $\ln 1.25 \approx 0.2231$; their sum is $\approx 0.0953 = \ln 1.1$. On the log scale the cancellation is visible directly: -0.2231 and $+0.2231$ add to zero.
3. False. Take $n = 2$ with $a_1 = a_2 = 1$: the left side is $\ln 2 \approx 0.693$, and the right side is $\ln 1 + \ln 1 = 0$. The logarithm turns a *product* into a sum; it does nothing so convenient to a sum.

3 Finite Sums with Closed Forms

A sum written in sigma notation is a set of instructions: do this addition, then that one, and so on. A **closed form** is a single formula that gives the same total directly, with no loop. For a sum with a million terms, the difference is a million additions versus one short calculation.

Speed is not the only reward. A closed form also tells you *how the total grows* as the number of terms grows, which a loop never shows you. We will see, for instance, that comparing every pair among n patients takes about $n^2/2$ comparisons, a fact that decides whether a computation on a large data set finishes in seconds or in days.

3.1 Gauss's Trick

The classic story: set the busywork of adding 1 through 100, a young Carl Friedrich Gauss wrote the answer down immediately. He had noticed that the list can be folded in half.

Here is the trick for a general stopping point n . Write the sum forward, then write it again backward underneath, and add the two rows column by column:

$$\begin{array}{rcccccc} S & = & 1 & + & 2 & + & 3 & + \cdots + & n \\ S & = & n & + & (n-1) & + & (n-2) & + \cdots + & 1 \\ \hline 2S & = & (n+1) & + & (n+1) & + & (n+1) & + \cdots + & (n+1) \end{array}$$

Every column adds up to the same thing. In the first column $1 + n = n + 1$; in the second, $2 + (n - 1) = n + 1$; in the third, $3 + (n - 2) = n + 1$. As you move one column to the right, the top number goes up by 1 and the bottom number goes down by 1, so their total never changes. There are n columns, each equal to $n + 1$, so

$$2S = \underbrace{(n+1) + (n+1) + \cdots + (n+1)}_{n \text{ columns, each equal to } n+1} = n(n+1).$$

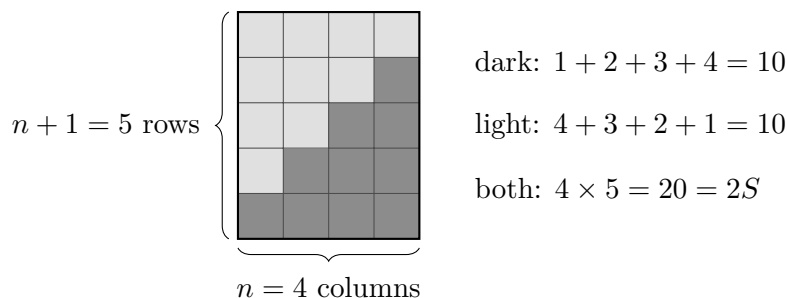
Divide by 2 to get the closed form.

Closed form: the sum of the first n counting numbers

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$

For Gauss's 1 through 100, that is $\frac{100 \cdot 101}{2} = 5050$: one multiplication and one division in place of ninety-nine additions.

The same argument can be drawn. Stack 1, 2, 3, 4 squares into a staircase (dark), then place a second copy of the staircase, turned upside down, on top of it (light). Together they fill a rectangle 4 squares wide and 5 squares tall:



The two staircases are the two rows of Gauss's table, and each column of the rectangle is one of his columns, always $n + 1$ tall. Half the rectangle is one staircase: $S = \frac{4 \times 5}{2} = 10$.

Before trusting a formula, it is good practice to test it on a few small cases you can add by hand:

n	Sum by hand	$\frac{n(n+1)}{2}$	Agree?
1	1	1	yes
2	$1 + 2 = 3$	3	yes
3	$1 + 2 + 3 = 6$	6	yes
4	$1 + 2 + 3 + 4 = 10$	10	yes
5	$1 + 2 + 3 + 4 + 5 = 15$	15	yes

A few checks do not prove a formula (the folding argument does that), but they catch a misremembered one immediately, for example $\frac{n(n-1)}{2}$, which gives 0 for $n = 1$.

Watch out: The formula expects a sum from 1, in steps of 1

$\frac{n(n+1)}{2}$ is the total of $1 + 2 + \cdots + n$ and nothing else. For a sum that starts somewhere else, or goes up in steps other than 1, first rewrite it in terms of $\sum i$ (by subtracting, pulling out a constant, or splitting), or use the folding argument directly. The examples below show all three.

Example 3.1: A different stopping point. $1 + 2 + \cdots + 50 = \frac{50 \cdot 51}{2} = 1275$.

Example 3.2: A sum that does not start at 1. Find $\sum_{i=11}^{50} i = 11 + 12 + \cdots + 50$. The idea is to take the full sum from 1 to 50 and remove the part you do not want, the terms 1 through 10:

$$\sum_{i=11}^{50} i = \sum_{i=1}^{50} i - \sum_{i=1}^{10} i = 1275 - \frac{10 \cdot 11}{2} = 1275 - 55 = 1220.$$

Notice that we subtract the sum up to 10, not up to 11: the term 11 belongs to the sum we want, so it must stay in. As a check, fold the list directly. It has $50 - 11 + 1 = 40$ terms, and folding pairs 11 with 50, 12 with 49, and so on, each pair totaling 61. Forty terms make 20 pairs, so the total is $20 \times 61 = 1220$. The two methods agree.

That check points to something general. Folding works for *any* list that goes up in equal steps, because moving one step inward from both ends always raises one number and lowers the other by the same amount. So

$$\text{total of an evenly spaced list} = (\text{number of terms}) \times \frac{\text{first} + \text{last}}{2},$$

that is, the number of terms times the average of the two ends. For $1 + 2 + \cdots + n$ this is $n \cdot \frac{1+n}{2}$, the closed form again.

Example 3.3: Steps of 5. Find $5 + 10 + 15 + \cdots + 100$. Two routes:

- **Pull out the constant.** Every term is 5 times a counting number, so the sum is $\sum_{i=1}^{20} 5i = 5 \sum_{i=1}^{20} i = 5 \cdot \frac{20 \cdot 21}{2} = 5 \cdot 210 = 1050$.
- **Fold.** There are 20 terms, and the ends average $\frac{5+100}{2} = 52.5$, so the total is $20 \times 52.5 = 1050$.

Having two independent routes to the same answer is the best check there is.

Example 3.4: The odd numbers. Find a closed form for $\sum_{i=1}^n (2i-1) = 1+3+5+\cdots+(2n-1)$. Split the sum and use the rules, then Gauss:

$$\sum_{i=1}^n (2i-1) = 2 \sum_{i=1}^n i - \sum_{i=1}^n 1 = 2 \cdot \frac{n(n+1)}{2} - n = n^2 + n - n = n^2.$$

The sum of the first n odd numbers is always a perfect square. For $n = 5$ that is 25, matching the direct addition $1+3+5+7+9$ in Section 2.2. There is a picture for this one too: each odd number is an L-shaped band that wraps around the square before it, turning a $k-1$ by $k-1$ square into a k by k square.



$1 + 3 + 5 + 7 + 9 = 5^2 = 25$ squares:
each band holds the next odd number.

(The notebook's section 2 checks this formula, and the one for the even numbers, against a loop.)

Example 3.5: Enrollment in a study. A clinical study enrolls 3 patients in week 1, 5 in week 2, 7 in week 3, and so on, two more each week. How many patients are enrolled after 12 weeks? Week i enrolls $2i + 1$ patients (check: week 1 gives 3), so the total is

$$\sum_{i=1}^{12} (2i+1) = 2 \sum_{i=1}^{12} i + 12 = 2 \cdot \frac{12 \cdot 13}{2} + 12 = 156 + 12 = 168 \text{ patients.}$$

Check by folding: week 12 enrolls $2(12) + 1 = 25$, and $12 \times \frac{3+25}{2} = 12 \times 14 = 168$.

Example 3.6: Comparing every pair of patients. To find duplicate records, a data team compares every pair of patients in a registry once. With n patients, how many comparisons is that?

Line the patients up. Patient 2 must be compared with the 1 patient before it, patient 3 with the 2 before it, and so on, up to patient n , compared with the $n - 1$ before it. The total is

$$1 + 2 + \cdots + (n - 1) = \sum_{i=1}^{n-1} i = \frac{(n - 1)n}{2},$$

using Gauss's formula with $n - 1$ in place of n (the sum stops at $n - 1$, so that is the “ n ” the formula needs). For 20 patients that is $\frac{19 \cdot 20}{2} = 190$ comparisons. For 1000 patients it is $\frac{999 \cdot 1000}{2} = 499,500$. The closed form shows what the loop hides: the count grows like $n^2/2$, so doubling the size of the registry roughly *quadruples* the work. This is why all-pairs computations are among the first things to become slow on large data sets.

3.2 The Finite Geometric Sum

A **geometric** sequence multiplies by the same **ratio** r at every step: $a, ar, ar^2, ar^3, \dots$. The first term is a , and each term is the one before it times r . The dosing sum from Section 1 is exactly of this form, with first term $a = 100$ and ratio $r = 0.5$.

To test whether a list is geometric, divide each term by the one before it. If you get the same number every time, the list is geometric and that number is r :

List	Ratios of neighbors	Geometric?
2, 6, 18, 54	3, 3, 3	yes, $r = 3$
100, 50, 25, 12.5	0.5, 0.5, 0.5	yes, $r = 0.5$ (the dosing sum)
$1, -\frac{1}{2}, \frac{1}{4}, -\frac{1}{8}$	$-\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}$	yes, $r = -\frac{1}{2}$ (signs alternate)
1, 2, 3, 4	2, 1.5, 1.33...	no: it <i>adds</i> 1 each step (Gauss's kind)
1, 4, 9, 16	4, 2.25, 1.78...	no: the squares

The fourth row is the one to be careful with. A list that goes up by a fixed *amount* (an arithmetic list) and a list that goes up by a fixed *factor* (a geometric list) behave completely differently, and they need different formulas.

A geometric sum has a closed form too, found by a trick as slick as Gauss's. Write the sum S_n of the first n terms, multiply the whole thing by r , and line the two rows up so that matching powers of r sit in the same column:

$$\begin{array}{rcl} S_n & = & a + ar + ar^2 + \cdots + ar^{n-1} \\ rS_n & = & ar + ar^2 + \cdots + ar^{n-1} + ar^n \\ \hline S_n - rS_n & = & a - ar^n \end{array}$$

Multiplying by r pushes every term one step along: a becomes ar , ar becomes ar^2 , and so on, with ar^{n-1} becoming the new term ar^n . So the two rows share every term from ar to ar^{n-1} , and subtracting the second row from the first cancels all of them. Only the first term of the top row and the last term of the bottom row survive.

What is left is easy to solve for S_n : factor S_n out of the left side, a out of the right, and divide by $1 - r$:

$$S_n(1 - r) = a(1 - r^n) \quad \implies \quad S_n = a \frac{1 - r^n}{1 - r}.$$

Closed form: the finite geometric sum

$$\sum_{i=0}^{n-1} ar^i = a \frac{1 - r^n}{1 - r} \quad (r \neq 1)$$

Two details are worth noticing. First, the sum starts at $i = 0$, so the exponent *is* the index, and running from 0 to $n - 1$ gives exactly n terms. Second, the condition $r \neq 1$ is there because the last step divides by $1 - r$, which is zero when $r = 1$. That case needs no formula anyway: every term is then just a , and a constant term counts itself, so the sum is na .

When $r > 1$, the formula produces a negative over a negative. Multiplying the top and the bottom by -1 gives an equivalent form with no minus signs, which many people find easier to use for growing lists:

$$a \frac{1 - r^n}{1 - r} = a \frac{r^n - 1}{r - 1}.$$

It is the same formula, not a new one, so use whichever you like.

Procedure: evaluating a finite geometric sum

1. a is the **first term actually in the sum** (write it out to be sure).
2. r is the ratio: any term divided by the one before it.
3. n is the **number of terms**. For $\sum_{i=m}^k$, that is $k - m + 1$.
4. If $r = 1$, the sum is na . Otherwise substitute into $a \frac{1 - r^n}{1 - r}$.

Example 3.7: A doubling outbreak. In a simple model of an outbreak, 1 person is infected in generation 1, and each generation of cases is twice the size of the one before: 1, 2, 4, 8, ... How many people have been infected in total after 10 generations? The sum is $1 + 2 + 4 + \cdots + 2^9$, with $a = 1$, $r = 2$, and $n = 10$ terms (the exponents run from 0 to 9). So

$$\sum_{i=0}^9 2^i = 1 \cdot \frac{2^{10} - 1}{2 - 1} = 1024 - 1 = 1023.$$

Notice that the finite formula works perfectly well with $r > 1$: a finite sum always has a total. (It is only the *infinite* version, in the next section, that needs r to be small.) Notice also that the total, 1023, is just one less than the next generation alone, $2^{10} = 1024$: in a doubling process, each new generation outnumbers all the earlier ones combined.

Example 3.8: Five doses. Check the table in Section 1: after 5 doses of 100 mg with half the drug surviving each interval, the amount is $\sum_{i=0}^4 100(0.5)^i$, with $a = 100$, $r = 0.5$, $n = 5$:

$$100 \cdot \frac{1 - 0.5^5}{1 - 0.5} = 100 \cdot \frac{1 - \frac{1}{32}}{\frac{1}{2}} = 200 \cdot \frac{31}{32} = 193.75 \text{ mg},$$

exactly the table's entry for dose 5. The formula did in one line what the table did in five additions, and it will do 500 doses just as fast.

Example 3.9: Finding the number of terms. Find $3 + 6 + 12 + \cdots + 384$. Here $a = 3$ and $r = 2$, but the number of terms is not given; we have to work it out from the last term. The terms

are $3 \cdot 2^0, 3 \cdot 2^1, 3 \cdot 2^2, \dots$, and $384 = 3 \times 128 = 3 \cdot 2^7$. So the exponents run from 0 to 7, which is $n = 8$ terms:

$$3 \cdot \frac{2^8 - 1}{2 - 1} = 3 \cdot 255 = 765.$$

Example 3.10: A negative ratio. Find $1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \frac{1}{16}$. The ratio is $r = -\frac{1}{2}$ (each term is the previous one times $-\frac{1}{2}$), with $a = 1$ and $n = 5$:

$$\frac{1 - \left(-\frac{1}{2}\right)^5}{1 - \left(-\frac{1}{2}\right)} = \frac{1 + \frac{1}{32}}{\frac{3}{2}} = \frac{33}{32} \cdot \frac{2}{3} = \frac{11}{16} = 0.6875.$$

Careful with the signs: $\left(-\frac{1}{2}\right)^5 = -\frac{1}{32}$, because an odd power of a negative number is negative, and subtracting it adds $\frac{1}{32}$. Check directly over a denominator of 16: $\frac{16-8+4-2+1}{16} = \frac{11}{16}$.

Example 3.11: A sum that starts later. Find $\sum_{i=3}^8 2^i$. Write out the first term to find a : it is $2^3 = 8$, not 1. The ratio is 2, and the number of terms is $8 - 3 + 1 = 6$. So

$$\sum_{i=3}^8 2^i = 8 \cdot \frac{2^6 - 1}{2 - 1} = 8 \cdot 63 = 504.$$

Check: $8 + 16 + 32 + 64 + 128 + 256 = 504$.

Example 3.12: Ratio 1. Find $7 + 7 + 7 + 7 + 7$. The ratio is $r = 1$, so the formula does not apply (it would divide by zero). A constant term counts itself: the total is $5 \times 7 = 35$.

Watch out: n counts terms; it is not the last exponent

In $a \frac{1-r^n}{1-r}$, the exponent n is the **number of terms**. For $\sum_{i=0}^9 2^i$ that is $n = 10$, even though the last exponent is 9. Using 2^9 by mistake gives $\frac{2^9-1}{1} = 511$ instead of 1023, a total that is off by exactly the last term, 512. A quick sanity check: for $n = 1$ the formula must return just a , and $a \frac{1-r}{1-r} = a$ does.

3.3 Practice: Closed Forms

Exercise 3.1

Use Gauss's formula, the rules for sums, or folding.

- $1 + 2 + 3 + \dots + 200$.
- $2 + 4 + 6 + \dots + 100$.
- $\sum_{i=21}^{40} i$.
- A study enrolls 4 patients in week 1, 7 in week 2, 10 in week 3, three more each week. How many are enrolled after 10 weeks?

Solution to Exercise 3.1

1. $\frac{200 \cdot 201}{2} = 20,100.$

2. Every term is 2 times a counting number, from $2 \cdot 1$ to $2 \cdot 50$, so pull out the 2:

$$\sum_{i=1}^{50} 2i = 2 \cdot \frac{50 \cdot 51}{2} = 2550.$$

Check by folding: 50 terms, with ends averaging $\frac{2+100}{2} = 51$, gives $50 \times 51 = 2550$.

3. Subtract the part you do not want (the terms 1 through 20, keeping 21):

$$\sum_{i=21}^{40} i = \frac{40 \cdot 41}{2} - \frac{20 \cdot 21}{2} = 820 - 210 = 610.$$

Check by folding: $40 - 21 + 1 = 20$ terms, ends averaging $\frac{21+40}{2} = 30.5$, gives $20 \times 30.5 = 610$.

4. Week i enrolls $3i + 1$ patients (check: $3(1) + 1 = 4$, $3(2) + 1 = 7$). So the total is

$$\sum_{i=1}^{10} (3i + 1) = 3 \cdot \frac{10 \cdot 11}{2} + 10 = 165 + 10 = 175 \text{ patients.}$$

Check by folding: week 10 enrolls 31, and $10 \times \frac{4+31}{2} = 10 \times 17.5 = 175$.

Exercise 3.2

Use the finite geometric sum. Identify a , r and n first.

1. $\sum_{i=0}^6 3 \cdot 2^i.$

2. $81 + 27 + 9 + 3 + 1.$

3. In a simple outbreak model, 5 people are infected in the first generation, and each generation is three times the size of the one before. How many people are infected in total over the first 6 generations?

4. Explain why the formula cannot be used for $4 + 4 + 4 + 4 + 4$, and find the sum anyway.

Solution to Exercise 3.2

1. $a = 3$ (the $i = 0$ term), $r = 2$, and the index runs from 0 to 6, so $n = 7$:

$$3 \cdot \frac{2^7 - 1}{2 - 1} = 3 \cdot 127 = 381.$$

2. $a = 81$, and each term is a third of the one before, so $r = \frac{1}{3}$, with $n = 5$ terms:

$$81 \cdot \frac{1 - \left(\frac{1}{3}\right)^5}{1 - \frac{1}{3}} = 81 \cdot \frac{1 - \frac{1}{243}}{\frac{2}{3}} = 81 \cdot \frac{242}{243} \cdot \frac{3}{2} = 121.$$

Check directly: $81 + 27 + 9 + 3 + 1 = 121$.

3. The generations are 5, 15, 45, 135, 405, 1215: $a = 5$, $r = 3$, $n = 6$. So

$$5 \cdot \frac{3^6 - 1}{3 - 1} = 5 \cdot \frac{728}{2} = 1820 \text{ people.}$$

4. Each term is the previous one times 1, so $r = 1$, and the formula would divide by $1 - r = 0$. No formula is needed: five copies of 4 make 20.

4 Infinite Series and Convergence

4.1 Partial Sums: How to Add Infinitely Many Things

The dosing question asks what happens if the doses go on forever, so it asks about a sum with infinitely many terms. That seems impossible: no person or computer can finish infinitely many additions. The way out is that we never actually add infinitely many numbers. We add the first n , and watch what happens to the total as n grows.

Definition: Partial sums and convergence

The n th **partial sum** is the total of the first n terms,

$$S_n = \sum_{i=1}^n a_i,$$

and these form a sequence of their own: $S_1, S_2, S_3, \dots$

The series $\sum_{i=1}^{\infty} a_i$ **converges** to S if those partial sums settle on S , that is, $S_n \rightarrow S$ as $n \rightarrow \infty$ (read “ S_n approaches S as n goes to infinity”). If they settle on no number, the series **diverges**.

So an infinite series is really a sequence in disguise: the sequence of running totals. A series can fail to settle in more than one way. The partial sums might grow past every number you could name, or they might keep bouncing back and forth without ever closing in on one value. Either way, it diverges.

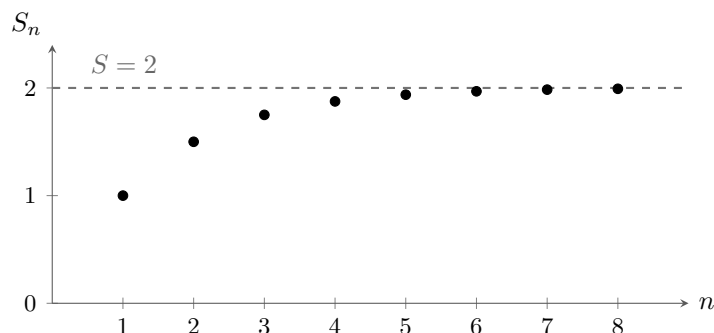
Partial sums are not an abstract device invented for infinite sums; you have seen them on every public-health dashboard.

Example 4.1: Running totals of data. A county reports new cases of an infection on five consecutive days: 4, 7, 3, 9, 5. The partial sums are the *cumulative* case counts:

$$S_1 = 4, \quad S_2 = 4 + 7 = 11, \quad S_3 = 14, \quad S_4 = 23, \quad S_5 = 28.$$

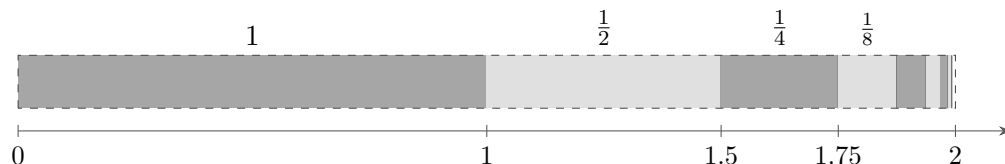
In code this is `np.cumsum(cases)`, or `df["cases"].cumsum()` in pandas. The “cumulative cases” curve on a dashboard is a plot of partial sums, and the question “is the outbreak leveling off?” is a question about whether they are settling down.

Here are the partial sums of $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$, where each term is half the one before:



The running totals are $S_1 = 1$, $S_2 = 1.5$, $S_3 = 1.75$, $S_4 = 1.875$, and so on. The gap between each total and 2 is 1, then 0.5, then 0.25, then 0.125: each new term is exactly half of the remaining gap, so it closes half the remaining gap to 2. The dots approach the dashed line without ever reaching it, and they get as close to it as you like if you go far enough.

The same thing can be seen as lengths laid end to end. A piece of length 1, then $\frac{1}{2}$, then $\frac{1}{4}$, and so on, each filling half of the room that is left before 2:



However many pieces you lay down, a sliver of room is always left, and the next piece fills exactly half of it. The pieces never pass 2, and the room left shrinks toward nothing.

That is exactly what “the sum is 2” means. **An infinite sum is a limit of ordinary finite additions**, not an addition: it is the number that the finite partial sums close in on.

What “settle on” means

“Settle on 2” has a precise meaning that you can test with numbers: *pick any tolerance, however small, and the partial sums eventually come within that tolerance of 2 and stay within it from then on*. For this series the gap after n terms is $2 - S_n = \left(\frac{1}{2}\right)^{n-1}$ (it was 1, 0.5, 0.25, ...), so we can see exactly when each tolerance is met:

Tolerance	First n with gap below it	Gap at that n
0.1	5	0.0625
0.01	8	0.0078125
0.001	11	0.0009765625

A tighter tolerance needs more terms, but every tolerance is eventually met, and because the gap only shrinks, it stays met. That is convergence. A series that diverges fails this test: for $1+1+1+\dots$ there is no number whose tolerance the partial sums ever stay within.

Example 4.2: Partial sums closing in on one half. Consider $\sum_{i=1}^{\infty} \frac{1}{3^i} = \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \dots$. Its first five partial sums are

$$\begin{aligned} S_1 &= \frac{1}{3} \approx 0.3333, & S_2 &= \frac{4}{9} \approx 0.4444, & S_3 &= \frac{13}{27} \approx 0.4815, \\ S_4 &= \frac{40}{81} \approx 0.4938, & S_5 &= \frac{121}{243} \approx 0.4979. \end{aligned}$$

They seem to be closing in on $\frac{1}{2}$. The gaps confirm it: $\frac{1}{2} - S_n$ is $\frac{1}{6}, \frac{1}{18}, \frac{1}{54}, \frac{1}{162}, \frac{1}{486}$, each one a third of the one before. A gap that is cut to a third at every step goes to zero, so the series converges to $\frac{1}{2}$. (Exercise 4.1 below confirms this with a formula.)

Example 4.3: Three ways to diverge. Each of these fails to settle, in its own way.

- $1 + 1 + 1 + \dots$: the partial sums are $S_n = n$, which climb forever.
- $1 + 2 + 3 + \dots$: the partial sums are $S_n = \frac{n(n+1)}{2}$ by Gauss's formula, so $S_{100} = 5050$, $S_{1000} = 500,500$, and they pass every number you could name.
- $1 - 1 + 1 - 1 + \dots$: the partial sums are $1, 0, 1, 0, \dots$. They stay small, but they never close in on a single number, so this series diverges too. Staying bounded is not the same as converging.

Watch out: Two different limits

For $1 + \frac{1}{2} + \frac{1}{4} + \dots$, two different things approach two different numbers. The *terms* $1, \frac{1}{2}, \frac{1}{4}, \dots$ approach 0. The *partial sums* $1, 1.5, 1.75, \dots$ approach 2. When we say a **series** converges, we always mean its partial sums, and its value is the limit of the partial sums, never the limit of the terms.

4.2 The Infinite Geometric Series

Now we can answer the question for every geometric series at once. Start from the finite formula of Section 3 and let n grow:

$$S_n = a \frac{1 - r^n}{1 - r}.$$

The only piece of this formula that changes with n is the single term r^n , so everything hinges on it. Here $|r|$ means the **absolute value** of r , its size with the sign ignored, so $|r| < 1$ says r lies strictly between -1 and 1 . When $|r| < 1$, repeated multiplication by r drives $r^n \rightarrow 0$: with $r = 0.5$, for instance, r^n runs $0.5, 0.25, 0.125, \dots$ toward zero. Then $1 - r^n \rightarrow 1$, and the formula collapses.

It is worth seeing how differently r^n behaves for a few ratios:

r	r^1	r^5	r^{10}	r^{50}	Behavior
0.5	0.5	0.03125	0.00098	8.9×10^{-16}	shrinks fast
0.9	0.9	0.59	0.35	0.0052	shrinks, but slowly
-0.5	-0.5	-0.03125	0.00098	8.9×10^{-16}	shrinks, sign alternating
1.1	1.1	1.61	2.59	117.4	grows without bound

With $r = 0.9$ the powers do head to zero, but after ten steps 0.9^{10} is still about 0.35. A ratio close to 1 converges, but slowly; we will see what that means for how many terms (or doses) it takes to get close.

The infinite geometric series

$$\sum_{i=0}^{\infty} ar^i = \frac{a}{1-r} \quad \text{provided } |r| < 1$$

The size of the ratio decides everything:

Ratio	Terms do what?	The series
$ r < 1$	shrink toward 0	converges to $\frac{a}{1-r}$
$r = \pm 1$	stay the same size	diverges
$ r > 1$	grow without bound	diverges

The middle row covers both ways of failing to settle. With $r = 1$ the series is $a + a + a + \dots$, and the partial sums climb forever. With $r = -1$ it is $a - a + a - a + \dots$, and the partial sums bounce between a and 0 without ever choosing one.

Worth memorizing: the sum is the first term divided by one minus the ratio. **Always confirm** $|r| < 1$ **first**. Plugging $r = 2$ into the formula returns $\frac{a}{1-2} = \frac{a}{-1} = -a$, a negative total for a sum of positive terms. That is the formula's way of saying it was never valid there.

Procedure: summing an infinite geometric series

1. Write out the first two or three terms.
2. a is the first term you wrote (not necessarily 1, and not necessarily the $i = 0$ term).
3. r is the second term divided by the first.
4. Check $|r| < 1$. If not, stop: the series diverges and has no sum.
5. The sum is $\frac{a}{1-r}$.

Example 4.4: Why $0.999\dots = 1$. The endless decimal $0.999\dots$ means the series $0.9 + 0.09 + 0.009 + \dots$. The first term is $a = 0.9$ and each term is a tenth of the one before, so $r = 0.1$, and $|r| < 1$. So

$$0.9 + 0.09 + 0.009 + \dots = \frac{0.9}{1-0.1} = \frac{0.9}{0.9} = 1.$$

The partial sums $0.9, 0.99, 0.999, \dots$ fall short of 1 by $0.1, 0.01, 0.001, \dots$, gaps that shrink to nothing. Since an endless decimal *means* the limit of its partial sums, $0.999\dots$ is exactly 1, not “just below” it. The feeling that it must fall short comes from picturing the partial sums, each of which does fall short; the infinite sum is their limit, and the limit is 1. (The notebook's section 4 runs this sum.)

Example 4.5: A repeating decimal as a fraction. Write $0.121212\dots$ as a fraction. Break it into its repeating blocks:

$$0.121212\dots = 0.12 + 0.0012 + 0.000012 + \dots$$

Each block is the previous one divided by 100, so $a = 0.12$ and $r = 0.01$:

$$\frac{0.12}{1 - 0.01} = \frac{0.12}{0.99} = \frac{12}{99} = \frac{4}{33}.$$

Dividing 4 by 33 on a calculator gives $0.121212\dots$, as it should. The same method gives $0.333\dots = \frac{0.3}{0.9} = \frac{1}{3}$.

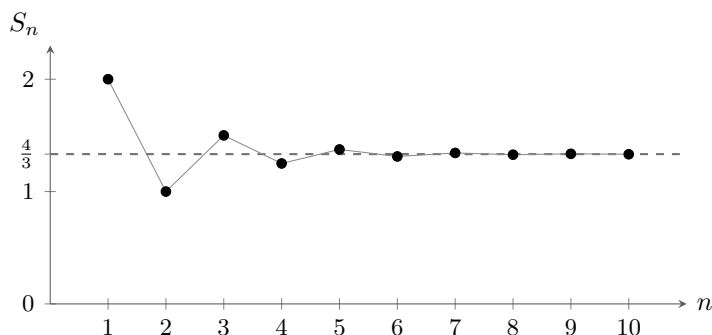
Example 4.6: A series that starts at a higher index. Find $\sum_{i=2}^{\infty} \left(\frac{2}{3}\right)^i$. Write out the first terms: $\left(\frac{2}{3}\right)^2 + \left(\frac{2}{3}\right)^3 + \dots = \frac{4}{9} + \frac{8}{27} + \dots$. So $a = \frac{4}{9}$ (not 1), and $r = \frac{2}{3}$, with $|r| < 1$:

$$\frac{4/9}{1 - 2/3} = \frac{4/9}{1/3} = \frac{4}{9} \cdot 3 = \frac{4}{3}.$$

Example 4.7: Signs that alternate. Find $2 - 1 + \frac{1}{2} - \frac{1}{4} + \dots$. Here $a = 2$ and $r = -\frac{1}{2}$ (each term is the previous one times $-\frac{1}{2}$), and $|r| = \frac{1}{2} < 1$:

$$\frac{2}{1 - \left(-\frac{1}{2}\right)} = \frac{2}{\frac{3}{2}} = \frac{4}{3}.$$

With a negative ratio the partial sums zigzag: 2, 1, 1.5, 1.25, 1.375, 1.3125, $\dots$, landing alternately above and below $\frac{4}{3}$ and closing in on it from both sides.



Example 4.8: An outbreak that burns out. In a simplified outbreak model, 100 people are infected in the first generation, and each generation of cases is 60% of the size of the one before: 100, 60, 36, 21.6, $\dots$ (Epidemiologists call such a multiplier a *reproduction number*; a value below 1 means each generation is smaller than the last.) The total number ever infected is

$$\sum_{i=0}^{\infty} 100(0.6)^i = \frac{100}{1 - 0.6} = \frac{100}{0.4} = 250.$$

Now change the multiplier to 0.9: the total becomes $\frac{100}{0.1} = 1000$. Raising the multiplier from 0.6 to 0.9 quadruples the size of the outbreak, because the factor $\frac{1}{1-r}$ grows very quickly as r approaches 1 ($\frac{1}{0.4} = 2.5$, but $\frac{1}{0.1} = 10$). And if the multiplier is 1 or more, the series diverges: in this model the outbreak never burns out on its own. The borderline $r = 1$ is exactly the threshold public-health officials watch.

Watch out: a is the first term of *this* sum

$\sum_{i=0}^{\infty} \left(\frac{1}{2}\right)^i = 2$, but $\sum_{i=1}^{\infty} \left(\frac{1}{2}\right)^i = \frac{1/2}{1-1/2} = 1$. The second sum is missing the $i = 0$ term, which is 1, so it is exactly 1 less. Always write out the first term before reading off a .

How close after n terms?

The two formulas together tell us exactly how far the n th partial sum is from the limit. Subtract:

$$\frac{a}{1-r} - a \frac{1-r^n}{1-r} = \frac{a - a + ar^n}{1-r} = \frac{a}{1-r} \cdot r^n.$$

In words: **after n terms, the remaining gap is the full sum times r^n .** Every additional term multiplies the gap by r . For $1 + \frac{1}{2} + \frac{1}{4} + \cdots$ the gap after n terms is $2 \cdot \left(\frac{1}{2}\right)^n = \left(\frac{1}{2}\right)^{n-1}$, which is the gap in the tolerance table of Section 4.1. For a ratio near 1 the picture is very different: with $r = 0.9$, after 10 terms the gap is still $0.9^{10} \approx 0.35$ of the full sum, so the partial sum has only reached about 65% of the limit.

One caution. Shrinking terms are *necessary* for a series to converge, but not *sufficient*. They are necessary because if the terms do not shrink toward zero, each new term keeps moving the total by a noticeable amount, so it cannot settle. They are not sufficient because the terms of

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots$$

shrink to zero, yet that sum grows without bound. Geometric shrinking, where each term is a fixed fraction of the one before, is fast enough; merely shrinking is not.

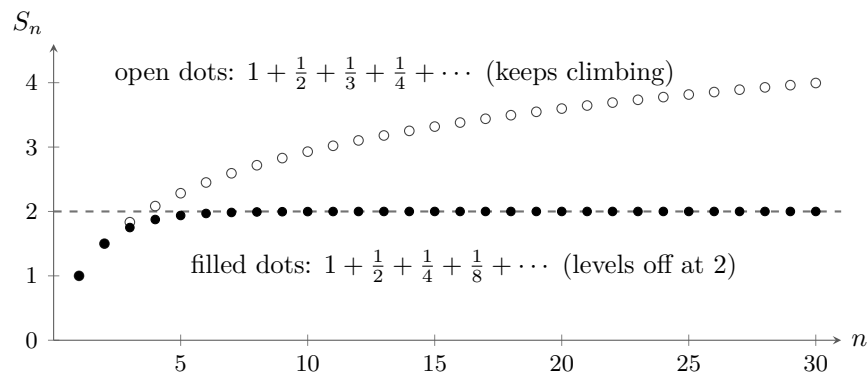
This series (called the **harmonic series**) is surprising enough to be worth a closer look. Its partial sums do grow, but extremely slowly:

Number of terms n	Partial sum $1 + \frac{1}{2} + \cdots + \frac{1}{n}$
10	2.929
100	5.187
1000	7.485
1,000,000	14.393

A million terms get the total only to about 14, and it takes 12,367 terms just to pass 10. If you only looked at a computer's output, you might well conclude that it levels off. It does not, and a grouping argument shows why. Group the terms in blocks that double in length:

$$1 + \frac{1}{2} + \underbrace{\left(\frac{1}{3} + \frac{1}{4}\right)}_{2 \text{ terms}} + \underbrace{\left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right)}_{4 \text{ terms}} + \underbrace{\left(\frac{1}{9} + \cdots + \frac{1}{16}\right)}_{8 \text{ terms}} + \cdots$$

In each block, every term is at least as big as the block's last term. So the first block is at least $2 \times \frac{1}{4} = \frac{1}{2}$, the second is at least $4 \times \frac{1}{8} = \frac{1}{2}$, the third at least $8 \times \frac{1}{16} = \frac{1}{2}$, and so on forever. There are infinitely many blocks, each adding at least $\frac{1}{2}$, so the total passes 1, then 2, then 100, then any number you name: the series diverges. The terms shrink, but not fast enough: the number of terms in each block grows exactly as fast as the terms shrink. Compare the two series side by side:



Both series start with the same two terms, and for a while the harmonic partial sums (open dots) look as if they might level off too. The geometric ones (filled dots) really do level off, at 2; the harmonic ones pass 2 at the fourth term, pass 3 at the eleventh, and keep going, slowly, forever.

4.3 Practice: Geometric Series

Exercise 4.1

Decide whether each series converges, and if it does, find its exact sum.

1. $\sum_{i=0}^{\infty} \frac{1}{2^i}$

2. $\sum_{i=1}^{\infty} \frac{1}{3^i}$

3. $\sum_{i=0}^{\infty} \left(\frac{3}{2}\right)^i$

4. $\sum_{i=0}^{\infty} 5 \left(-\frac{1}{4}\right)^i$

For each one, identify a and r *first*: most mistakes here are bookkeeping mistakes about which term is the first one.

Solution to Exercise 4.1

For each series, write out the first few terms: a is the first term written, and r is what each term is multiplied by to get the next one.

1. The terms are $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots$, so $a = 1$ (the $i = 0$ term, since $2^0 = 1$) and $r = \frac{1}{2}$. Since $|r| = \frac{1}{2} < 1$, it converges:

$$\frac{1}{1 - 1/2} = \frac{1}{1/2} = 2.$$

This is the series whose partial sums are plotted in Section 4.1, and the formula agrees with the dashed line there.

2. This sum starts at $i = 1$, so the first term is $a = \frac{1}{3}$, not 1. The terms are $\frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \cdots$,

so $r = \frac{1}{3}$, and $|r| < 1$:

$$\frac{1/3}{1 - 1/3} = \frac{1/3}{2/3} = \frac{1}{2}.$$

Had you taken $a = 1$ by mistake, you would have gotten $1/(1 - 1/3) = 3/2$, which is the sum *starting from* $i = 0$. The two answers differ by exactly 1, the $i = 0$ term that this series leaves out. The answer $\frac{1}{2}$ also confirms what the partial sums suggested in Section 4.1.

3. The terms are $1 + \frac{3}{2} + \frac{9}{4} + \dots$, so $a = 1$ and $r = \frac{3}{2} > 1$. The terms grow, so the series **diverges**, and no sum exists. The formula would have returned

$$\frac{1}{1 - 3/2} = \frac{1}{-1/2} = -2,$$

a negative answer for a sum of positive terms, which is nonsense, as promised.

4. The terms are $5 - \frac{5}{4} + \frac{5}{16} - \frac{5}{64} + \dots$, so $a = 5$ and $r = -\frac{1}{4}$. Here $|r| = \frac{1}{4} < 1$, so it converges even though the terms alternate in sign:

$$\frac{5}{1 - (-1/4)} = \frac{5}{1 + 1/4} = \frac{5}{5/4} = 4.$$

The partial sums 5, 3.75, 4.0625, 3.984375, ... land alternately above and below 4, closing in on it from both sides.

Exercise 4.2

Decide whether each converges; if it does, find its sum.

1. $\sum_{i=0}^{\infty} 3(0.4)^i$

2. $\sum_{i=1}^{\infty} (0.2)^i$

3. $\sum_{i=0}^{\infty} (-1.1)^i$

4. The repeating decimal $0.454545\dots$, written as a fraction.

Solution to Exercise 4.2

1. $a = 3$, $r = 0.4$, and $|r| < 1$: the sum is $\frac{3}{1 - 0.4} = \frac{3}{0.6} = 5$.

2. The first term is the $i = 1$ term, $a = 0.2$, and $r = 0.2$: the sum is $\frac{0.2}{1 - 0.2} = \frac{0.2}{0.8} = 0.25$.

3. $r = -1.1$, so $|r| = 1.1 > 1$: the terms $1, -1.1, 1.21, -1.331, \dots$ alternate in sign but grow in size, and the partial sums swing wider and wider. It diverges. (The formula would give

$\frac{1}{2.1}$, a number that means nothing here.)

4. $0.454545 \dots = 0.45 + 0.0045 + 0.000045 + \dots$, with $a = 0.45$ and $r = 0.01$:

$$\frac{0.45}{1 - 0.01} = \frac{0.45}{0.99} = \frac{45}{99} = \frac{5}{11}.$$

Exercise 4.3

The series $\sum_{i=0}^{\infty} \left(\frac{1}{3}\right)^i$ converges. What is its sum, and how many terms are needed before the partial sum is within 0.01 of it?

Solution to Exercise 4.3

Here $a = 1$ and $r = \frac{1}{3}$, so the sum is $\frac{1}{1-1/3} = \frac{3}{2} = 1.5$. After n terms the gap is the full sum times r^n , that is $1.5 \cdot \left(\frac{1}{3}\right)^n$. Try values of n :

$$n = 3 : 1.5 \cdot \frac{1}{27} \approx 0.0556, \quad n = 4 : 1.5 \cdot \frac{1}{81} \approx 0.0185, \quad n = 5 : 1.5 \cdot \frac{1}{243} \approx 0.0062.$$

So 5 terms are needed: $1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \frac{1}{81} = \frac{121}{81} \approx 1.4938$, which is within 0.0062 of 1.5, while four terms leave a gap of about 0.0185. With a ratio of $\frac{1}{3}$, every extra term cuts the gap to a third, so convergence is fast.

5 Back to the Dosing Problem

5.1 Answering the Opening Question

We now have the vocabulary to finish the problem from Section 1. Call the dose $D = 100$ mg, taken every 6 hours, and let $r = 0.5$ be the fraction of the drug that survives until the next dose. A dose taken k doses ago has been cut in half k times, so it contributes Dr^k . Just after the n th dose, the body holds

$$A_n = D + Dr + Dr^2 + \dots + Dr^{n-1} = D \frac{1 - r^n}{1 - r},$$

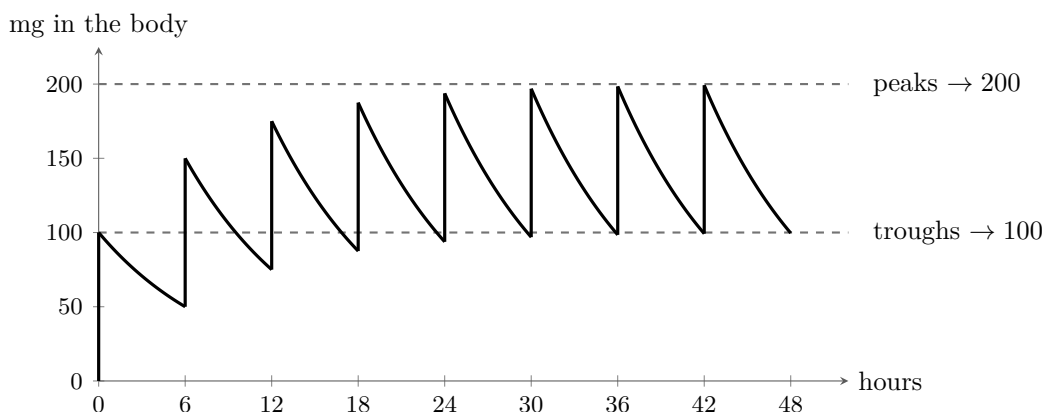
a **finite geometric sum**, exactly the one derived in Section 3, with first term D and ratio r . With these numbers, A_n is 100 times the partial sums plotted in Section 4.1: 100, then 150, then 175, then 187.5 mg, and so on.

Since $|r| < 1$, the term r^n dies away as the doses continue, and the amount levels off at a **steady state**, a level that the amount approaches and then holds. Writing A_{∞} for the value that A_n approaches as $n \rightarrow \infty$:

$$A_{\infty} = \frac{D}{1 - r} = \frac{100}{1 - 0.5} = 200 \text{ mg}.$$

The drug does accumulate, but only to **twice** a single dose, never without bound. Each new dose is eventually offset by what the body clears. You can check this at the steady state itself: 200 mg halves to 100 mg over the next 6 hours, and the next 100 mg dose brings it right back to 200. The sum of infinitely many doses is finite.

The amount in the body is not constant between doses: it jumps up by 100 mg at each dose and then decays until the next one. Plotting the amount against time over the first two days gives a sawtooth:



The top of each tooth is A_n , the amount just after dose n . The bottom of each tooth, just before the next dose, is what that amount has decayed to over one interval, rA_n ; this lowest level is called the **trough**. Both climb toward a ceiling: the peaks toward $A_\infty = 200$ mg, and the troughs toward $rA_\infty = 100$ mg. At steady state the amount swings between 100 and 200 mg every 6 hours. Both numbers matter clinically: the peak is often what matters for side effects, and the trough for whether the drug is still doing its job just before the next dose.

How quickly does the amount approach the steady state? Divide the two formulas:

$$\frac{A_n}{A_\infty} = \frac{D \frac{1-r^n}{1-r}}{\frac{D}{1-r}} = 1 - r^n.$$

So after n doses, the peak has reached the fraction $1 - r^n$ of its steady-state value, and the dose size D has canceled out: *how fast* the level approaches steady state depends only on r .

Dose n	Peak A_n (mg)	Next trough $0.5A_n$ (mg)	Share of steady state, $1 - 0.5^n$
1	100	50	50%
2	150	75	75%
3	175	87.5	87.5%
4	187.5	93.75	93.75%
5	193.75	96.875	96.875%
$\vdots$	$\vdots$	$\vdots$	$\vdots$
∞	200	100	100%

Example 5.1: How many doses to reach 90% of steady state. We need $1 - 0.5^n \geq 0.9$, that is, $0.5^n \leq 0.1$. Powers of 0.5 are 0.5, 0.25, 0.125, 0.0625, ..., so $n = 3$ is not enough ($0.125 > 0.1$), and $n = 4$ is ($0.0625 \leq 0.1$). After 4 doses, 24 hours in, the peak is 187.5 mg, which is 93.75% of the steady state. Since the dosing interval here equals the half-life, 4 doses is 4 half-lives, and 5 doses gets to 96.9%. This is the source of the clinical rule of thumb that a drug given regularly reaches steady state after about **four to five half-lives**.

5.2 More Dosing Scenarios

The same few formulas answer every version of the question. Collected in one place:

Repeated dosing: the formulas

A dose D is given at regular intervals, and a fraction r of what is in the body survives each interval ($0 < r < 1$). Then

- amount just after dose n (the **peak**): $A_n = D \frac{1 - r^n}{1 - r}$;
- **steady-state peak**: $A_\infty = \frac{D}{1 - r}$, and the steady-state **trough** is rA_∞ ;
- fraction of steady state reached after n doses: $1 - r^n$.

The number $\frac{1}{1-r}$ says how many single doses' worth of drug the body holds at steady state; it is sometimes called the **accumulation ratio**. For $r = 0.5$ it is 2.

Where r comes from. In a problem, r is usually not given directly; you get it from the half-life h and the dosing interval τ (the Greek letter tau). Each half-life cuts the amount in half, and an interval contains τ/h half-lives, so

$$r = \left(\frac{1}{2}\right)^{\tau/h}.$$

If the interval equals the half-life, $r = \frac{1}{2}$. If the interval is two half-lives (a 6-hour half-life, dosed every 12 hours), the amount is halved twice, so $r = \frac{1}{4}$. If the interval is half a half-life (a 12-hour half-life, dosed every 6 hours), $r = \left(\frac{1}{2}\right)^{1/2} \approx 0.7071$, using the fractional exponents of Unit 1-3.

Watch out: r is what *remains*, not what is cleared

If a problem says “the body clears 30% of the drug between doses,” then 70% remains, and $r = 0.7$, not 0.3. With $D = 100$ mg the steady state is $\frac{100}{0.3} \approx 333.3$ mg; using $r = 0.3$ by mistake gives $\frac{100}{0.7} \approx 142.9$ mg, badly underestimating how much drug builds up.

The worked examples below run from direct substitution to designing a regimen backward from a target.

Example 5.2: A bigger dose, a longer interval. A patient takes 250 mg every 8 hours, and the half-life is 8 hours. The interval equals the half-life, so $r = 0.5$. The first three peaks are

$$A_1 = 250, \quad A_2 = 250 + 125 = 375, \quad A_3 = 250 \frac{1 - 0.5^3}{0.5} = 500 \times 0.875 = 437.5 \text{ mg},$$

and the steady-state peak is $\frac{250}{1-0.5} = 500$ mg, with a trough of 250 mg. The steady state is again twice the dose: the accumulation ratio $\frac{1}{1-r}$ depends only on r , so any drug dosed once per half-life settles at twice its dose.

Example 5.3: Dosing less often than the half-life. A drug with a 6-hour half-life is given as 100 mg every 12 hours. Two half-lives pass between doses, so $r = \left(\frac{1}{2}\right)^2 = 0.25$. Then

$$A_1 = 100, \quad A_2 = 100 + 25 = 125, \quad A_3 = 100 + 25 + 6.25 = 131.25,$$

and the steady-state peak is

$$A_{\infty} = \frac{100}{1 - 0.25} = \frac{100}{0.75} \approx 133.3 \text{ mg.}$$

Very little accumulation happens, since most of each dose is gone before the next one. But the swings are large: the steady-state trough is $0.25 \times 133.3 \approx 33.3$ mg, so the level drops to a quarter of its peak before every dose. If the drug only works above, say, 50 mg, this regimen leaves the patient unprotected for part of every interval, even though the average level looks fine.

Example 5.4: A slowly cleared drug. A drug of which 80% remains between doses ($r = 0.8$) is given as 50 mg per dose. The steady-state peak is

$$A_{\infty} = \frac{50}{1 - 0.8} = \frac{50}{0.2} = 250 \text{ mg,}$$

five times a single dose. How many doses until the peak reaches 90% of that, 225 mg? We need $1 - 0.8^n \geq 0.9$, that is, $0.8^n \leq 0.1$. Tabulating powers of 0.8:

n	7	8	9	10	11	12
0.8^n	0.2097	0.1678	0.1342	0.1074	0.0859	0.0687
A_n (mg)	197.6	208.1	216.4	223.2	228.5	232.8

So 10 doses fall just short (223.2 mg), and it takes 11 doses. Compare the drug dosed once per half-life, which reached 90% in 4 doses: a ratio close to 1 means more accumulation *and* a slower approach, exactly the “converges, but slowly” behavior of $r = 0.9$ in Section 4.2.

Logarithms from Unit 1-3 give the same answer without a table. Take $\ln$ of both sides of $0.8^n \leq 0.1$ (the logarithm is increasing, so the inequality keeps its direction):

$$n \ln 0.8 \leq \ln 0.1 \quad \implies \quad n \geq \frac{\ln 0.1}{\ln 0.8} \approx \frac{-2.3026}{-0.2231} \approx 10.3,$$

so the first whole number of doses that works is $n = 11$.

Watch out: Dividing by a negative logarithm flips the inequality

For any r between 0 and 1, $\ln r$ is negative (here $\ln 0.8 \approx -0.2231$). Dividing both sides of an inequality by a negative number reverses it, which is why $n \ln 0.8 \leq \ln 0.1$ became $n \geq \dots$ above. Forgetting the flip gives “ $n \leq 10.3$,” which says the opposite of the truth. When in doubt, check your answer by plugging a value into r^n , as in the table.

Example 5.5: A half-life longer than the dosing interval. A drug with a 12-hour half-life is given as 100 mg every 6 hours. Only half a half-life passes between doses, so

$$r = \left(\frac{1}{2}\right)^{6/12} = \left(\frac{1}{2}\right)^{1/2} = \frac{1}{\sqrt{2}} \approx 0.7071.$$

Then $A_{\infty} = \frac{100}{1 - 0.7071} \approx \frac{100}{0.2929} \approx 341.4$ mg, about 3.4 doses’ worth, and the steady-state trough is $0.7071 \times 341.4 \approx 241.4$ mg. Compared with the opening problem (same dose and same interval, but twice the half-life), the steady state is about 70% higher: 341.4 mg against 200 mg. Doubling the

half-life did *not* double the steady state; the relationship runs through $\frac{1}{1-r}$, which is not proportional to anything simple.

Example 5.6: Designing the dose from a target. A clinician wants the steady-state peak to be 300 mg, for a drug dosed once per half-life ($r = 0.5$). What dose should be given? Solve $A_\infty = \frac{D}{1-r}$ for D :

$$D = A_\infty(1 - r) = 300 \times (1 - 0.5) = 150 \text{ mg.}$$

The steady-state trough is then $0.5 \times 300 = 150$ mg. Working backward from a target level, rather than forward from a dose, is how dosing regimens are actually designed, and it uses exactly the same formula read in the other direction.

5.3 Practice: Dosing

Exercise 5.1

A patient takes 400 mg every 12 hours; the half-life is 12 hours. Find the amount just after the third dose, the steady-state peak, and the steady-state trough.

Solution to Exercise 5.1

The interval equals the half-life, so $r = 0.5$. After the third dose,

$$A_3 = 400 + 200 + 100 = 700 \text{ mg} \quad \left(\text{or } 400 \frac{1-0.125}{0.5} = 800 \times 0.875 = 700 \right).$$

The steady-state peak is $\frac{400}{1-0.5} = 800$ mg, and the trough is $0.5 \times 800 = 400$ mg. As with every drug dosed once per half-life, the steady-state peak is twice the dose and the trough equals the dose.

Exercise 5.2

The body clears 60% of a drug between doses, and each dose is 90 mg.

1. Find r , and the amounts just after the first three doses.
2. Find the steady-state peak.
3. What dose would give a steady-state peak of 200 mg instead?

Solution to Exercise 5.2

1. 60% is cleared, so 40% remains: $r = 0.4$ (not 0.6). The amounts are $A_1 = 90$, $A_2 = 90 + 36 = 126$, and $A_3 = 90 + 36 + 14.4 = 140.4$ mg.
2. $A_\infty = \frac{90}{1-0.4} = \frac{90}{0.6} = 150$ mg. Already after three doses the peak is at $1 - 0.4^3 = 93.6\%$ of this, because a small r means fast convergence.
3. $D = A_\infty(1 - r) = 200 \times 0.6 = 120$ mg.

Exercise 5.3

For a drug dosed once per half-life ($r = 0.5$), how many doses does it take for the peak to reach 95% of its steady state?

Solution to Exercise 5.3

We need $1 - 0.5^n \geq 0.95$, that is, $0.5^n \leq 0.05$. Four doses give $0.5^4 = 0.0625$, not small enough; five give $0.5^5 = 0.03125$, which is. So it takes 5 doses, reaching 96.875% of steady state. (With logarithms: $n \geq \frac{\ln 0.05}{\ln 0.5} \approx 4.32$, remembering to flip the inequality, so $n = 5$.) This is the upper end of the “four to five half-lives” rule of thumb.

Exercise 5.4

Since it takes several doses to approach the steady state, clinicians sometimes start with a larger **loading dose**. Suppose the first dose is 200 mg, and every later dose is the usual 100 mg every 6 hours, with a 6-hour half-life. Find the amount just after each of the first four doses. What do you notice, and why does it happen?

Solution to Exercise 5.4

Just after dose 1 there are 200 mg. Six hours later that has halved to 100 mg, and the 100 mg dose brings it to 200 mg. The same thing happens every time:

dose 1 : 200, dose 2 : $100 + 100 = 200$, dose 3 : $100 + 100 = 200$, dose 4 : 200.

The peak is at the steady-state value 200 mg from the very first dose. It happens because the loading dose was chosen to be exactly $A_\infty = \frac{D}{1-r} = 200$ mg: at the steady state, what is cleared over one interval (100 mg) is exactly what one maintenance dose replaces. In series terms, just after dose n the loading dose contributes $200(0.5)^{n-1}$ and the later doses contribute a finite geometric sum,

$$200(0.5)^{n-1} + 100 \frac{1 - 0.5^{n-1}}{1 - 0.5} = 200(0.5)^{n-1} + 200 - 200(0.5)^{n-1} = 200,$$

so the two pieces trade off exactly. Clinically, a loading dose skips the four to five half-lives of waiting.

6 Review

What we covered

1. A **sequence** is a list; a **series** is its total. **Sigma notation** $\sum_{i=1}^n a_i$ is a for-loop written compactly: start, stop, and a rule for the term.
2. The index is a **dummy variable**, so its name never matters and it never escapes the sum.
3. Sums are **linear**: constants pull out and sums split. But products and squares do *not* pass through: $\sum a_i b_i \neq (\sum a_i)(\sum b_i)$.
4. **Closed forms** replace a loop with a formula: $\sum_{i=1}^n i = \frac{n(n+1)}{2}$, and the geometric sum $a \frac{1-r^n}{1-r}$.
5. An infinite series is a **limit of partial sums**: it converges if S_n settles on a number, and diverges otherwise.
6. **Geometric series**: $\sum_{i=0}^{\infty} ar^i = \frac{a}{1-r}$ when $|r| < 1$. This is the one to memorize, and the reason a repeated dose levels off instead of piling up forever.

Formulas at a glance

Number of terms in $\sum_{i=m}^n$	$n - m + 1$
Linearity	$\sum (c a_i + b_i) = c \sum a_i + \sum b_i, \quad \sum_{i=1}^n c = nc$
Gauss	$\sum_{i=1}^n i = \frac{n(n+1)}{2}$
Finite geometric (n terms, $r \neq 1$)	$a \frac{1-r^n}{1-r}$
Infinite geometric ($ r < 1$)	$\frac{a}{1-r}, \quad \text{gap after } n \text{ terms} = \frac{a}{1-r} r^n$
Repeated dosing	$A_n = D \frac{1-r^n}{1-r}, \quad A_{\infty} = \frac{D}{1-r}, \quad \frac{A_n}{A_{\infty}} = 1 - r^n$

Notebook: U1-4_Series-1_Convergence.ipynb

Previous: Unit 1-3, **Functions**.

Next: Unit 1-5, **Floating-Point Arithmetic**: how a computer actually stores numbers, and why a long product of small probabilities can collapse to zero.